Lecture 31

Greedy: Kruskal's Algorithm (contd.), Prim's Algorithm

Source: Introduction to Algorithms, CLRS

Kruskal's Algorithm: Correctness (Minimum)

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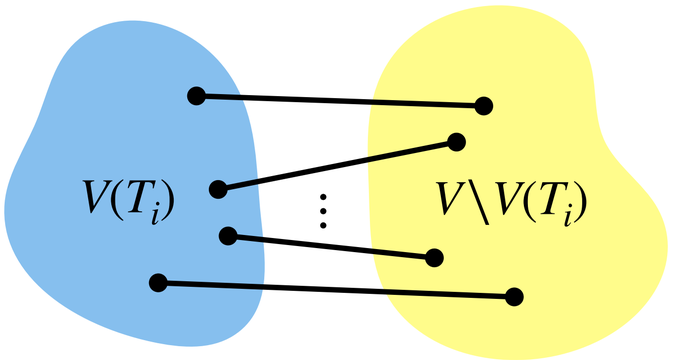

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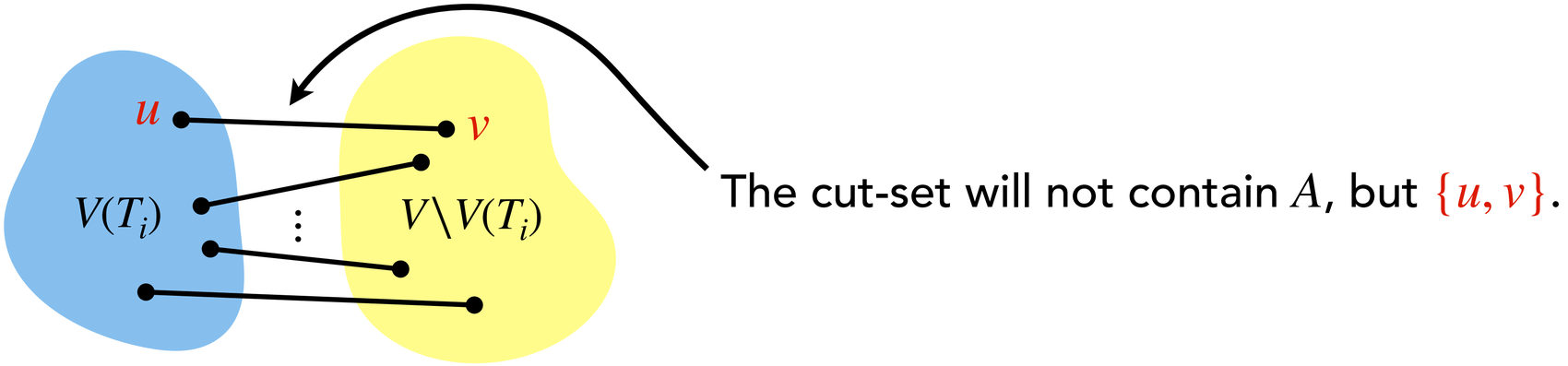

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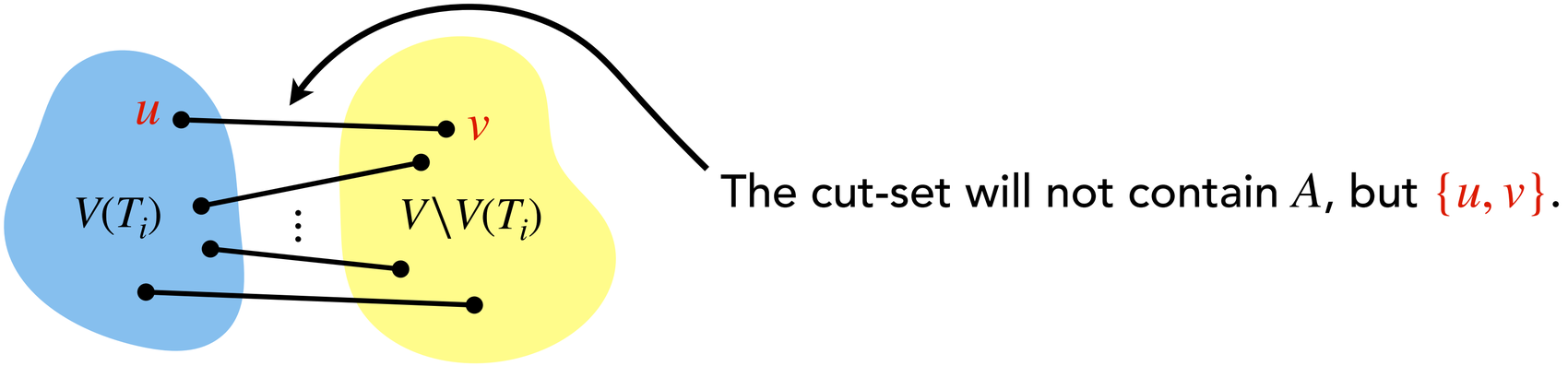

If $\{u, v\}$ is a least weight edge in the above cut-set, then we are done.

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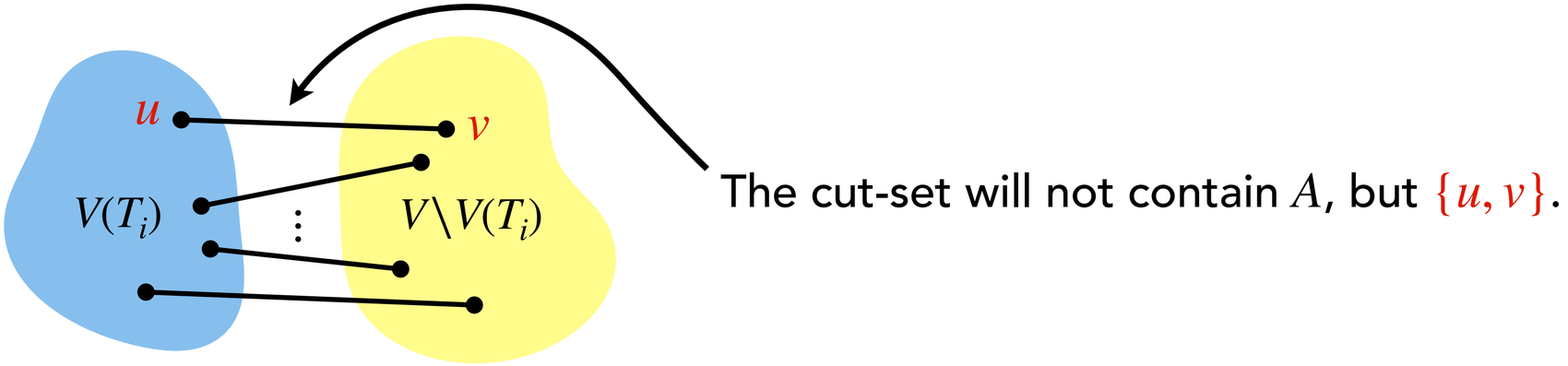

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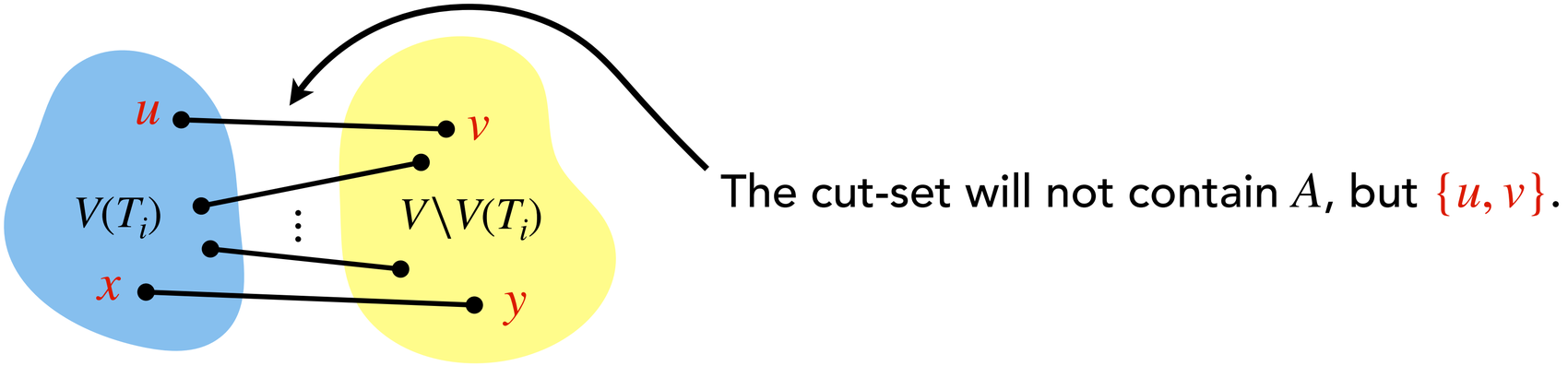

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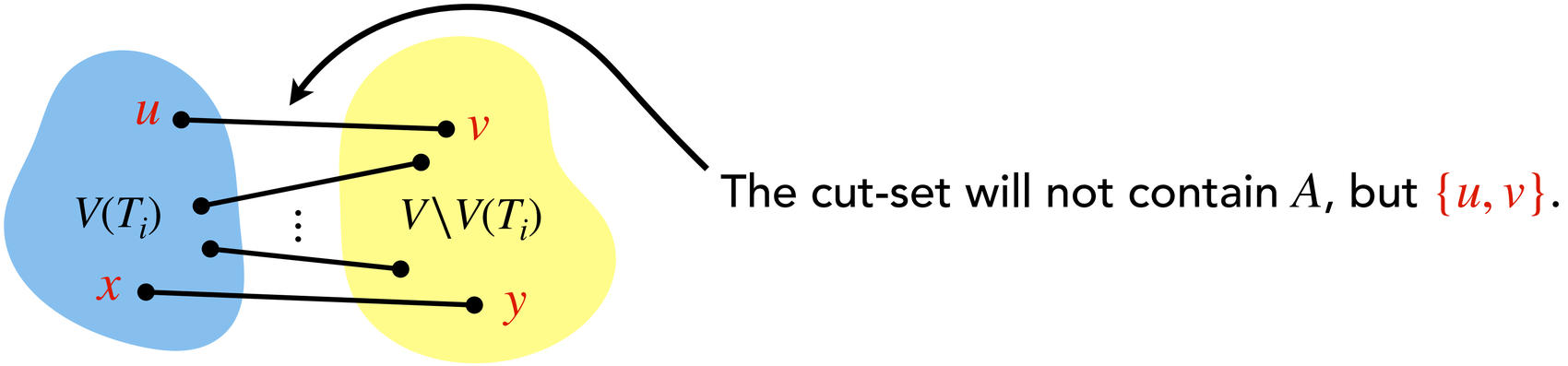

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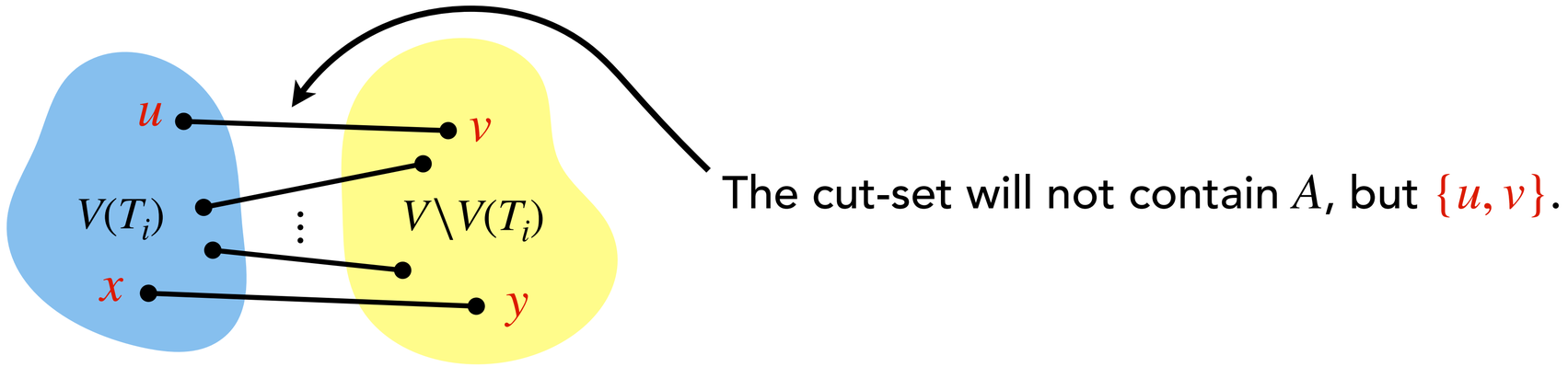

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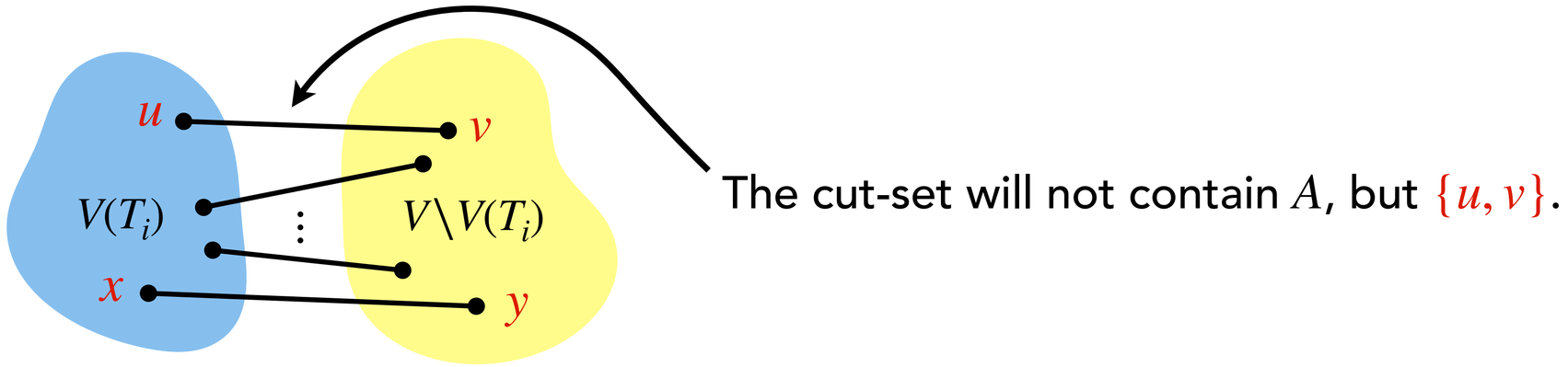

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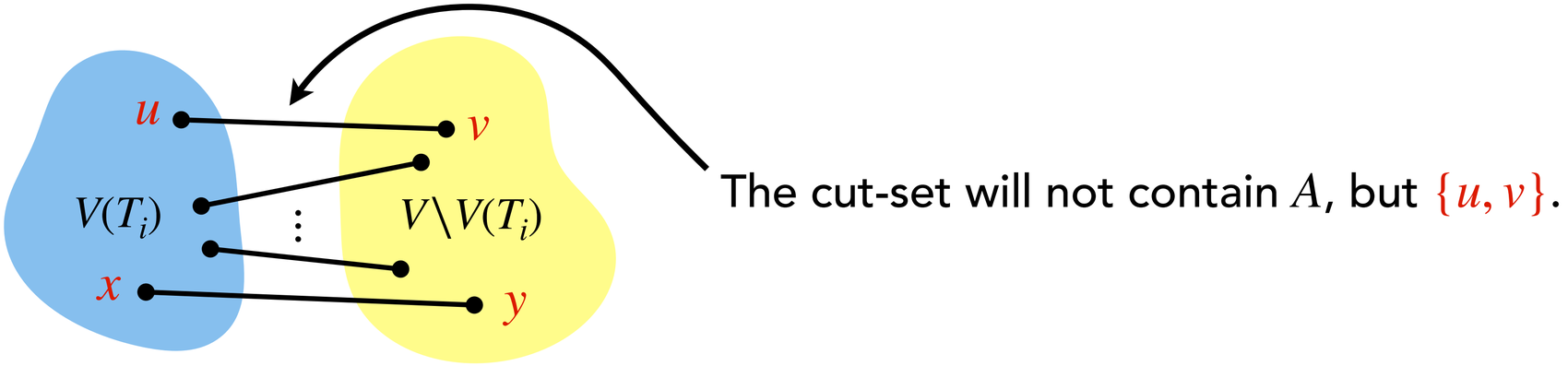

If $\{x, y\}$ not $\{u, v\}$ is a least weight edge of the cut-set, then algorithm would have considered $\{x, y\}$ before $\{u, v\}$and have added $\{x, y\}$ to A putting x and y in same subtree. Contradiction!

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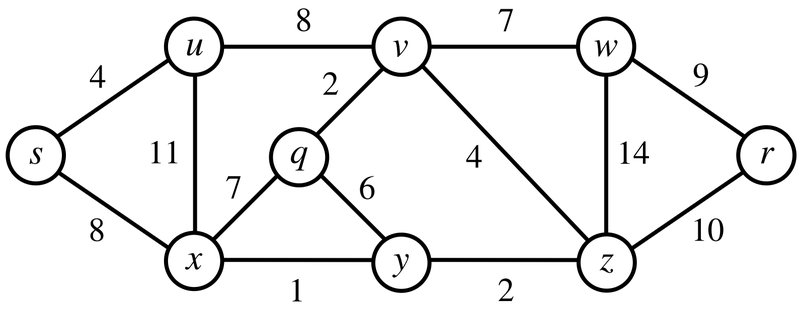

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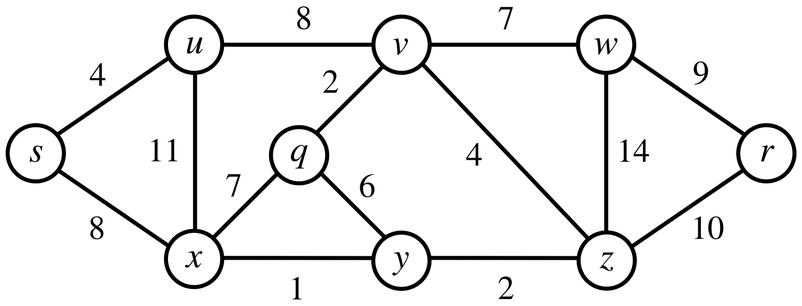

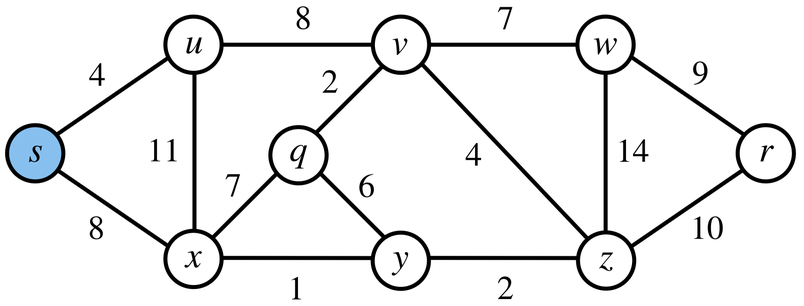

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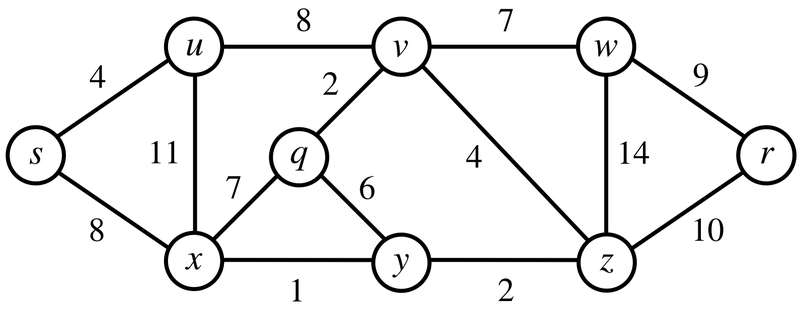

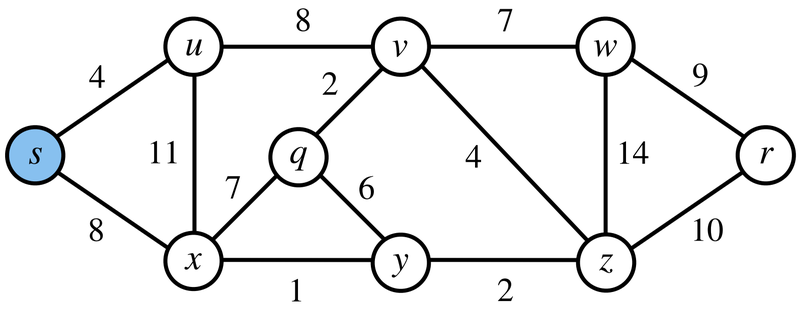

Build a "rooted" spanning tree step by step by repeatedly adding a least weight edge

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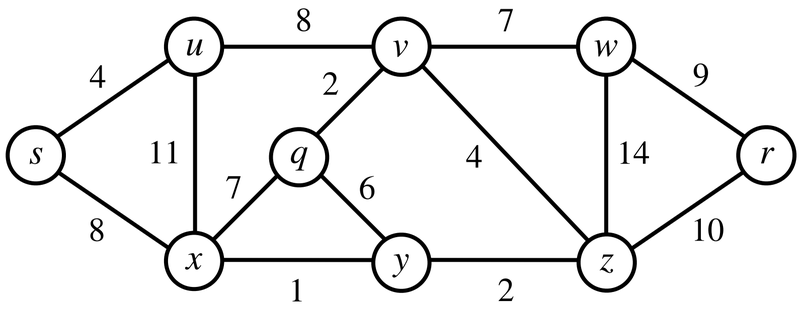

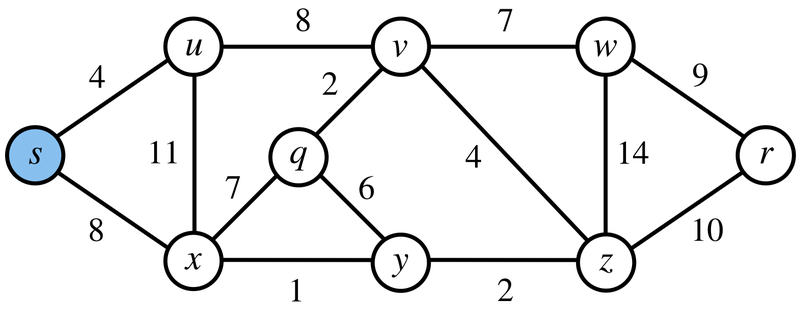

Build a "rooted" spanning tree step by step by repeatedly adding a least weight edge from the cut formed by the vertices of the tree and the remaining vertices.

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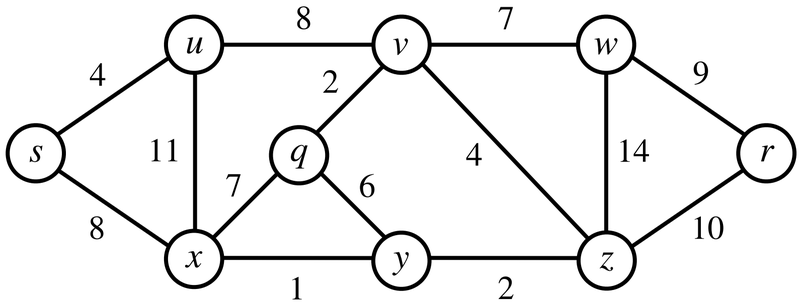

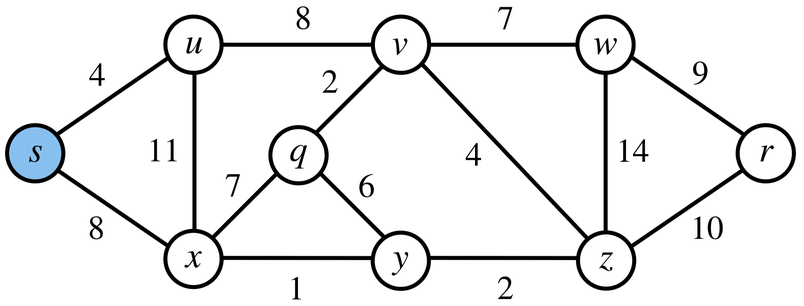

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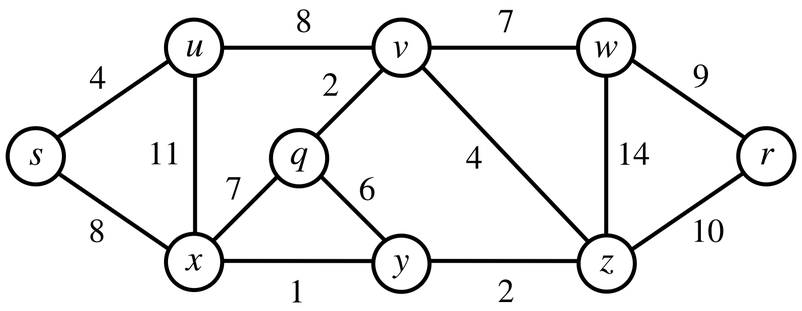

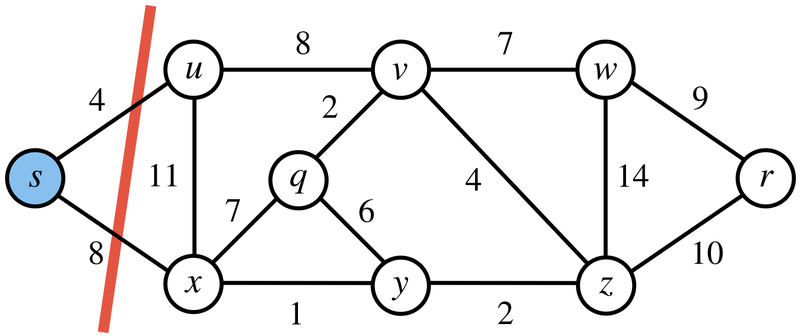

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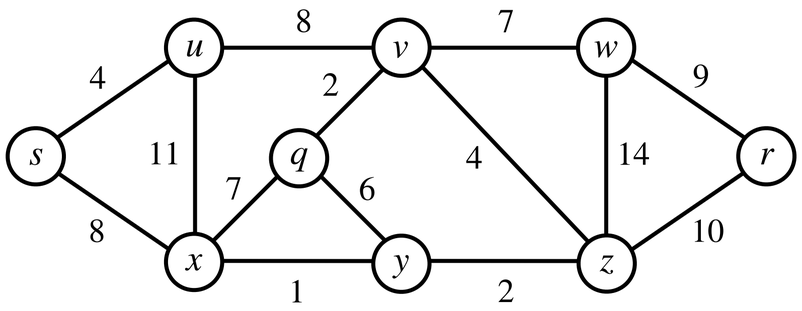

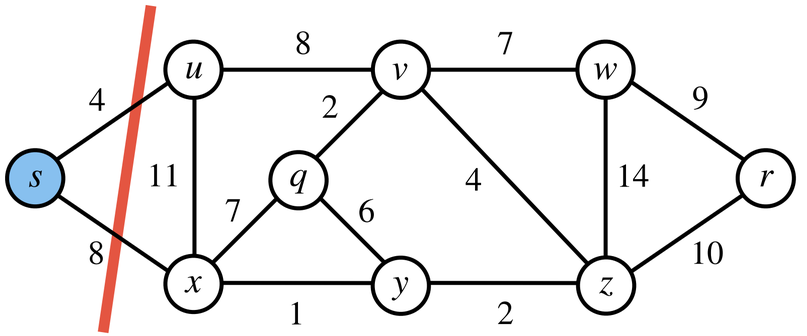

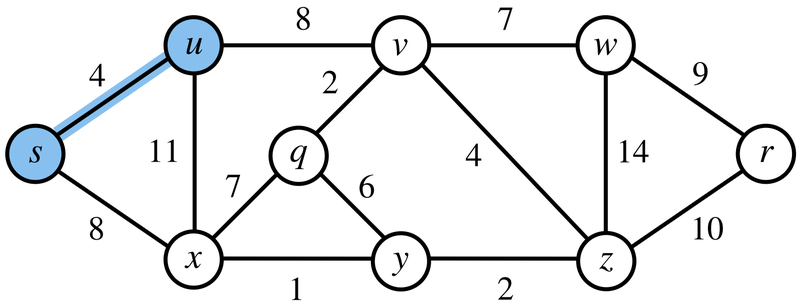

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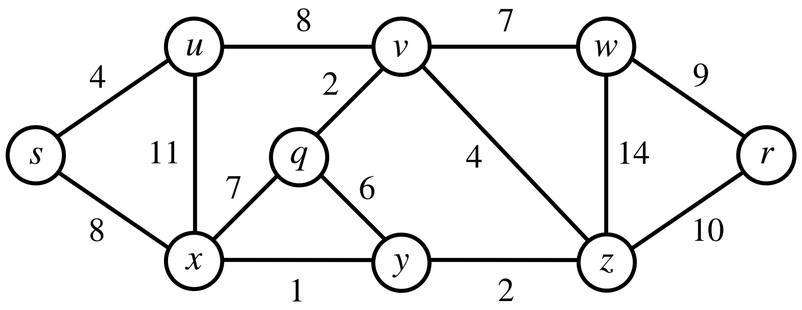

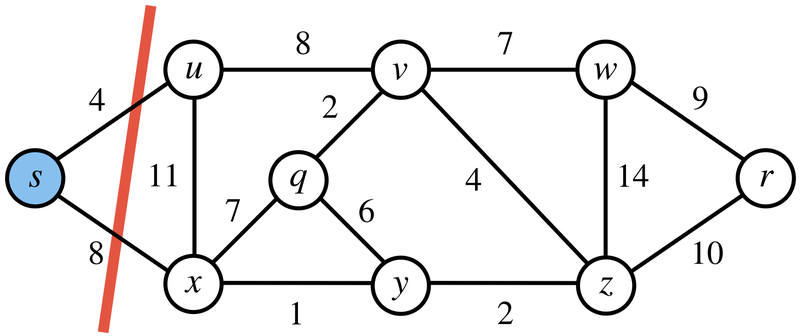

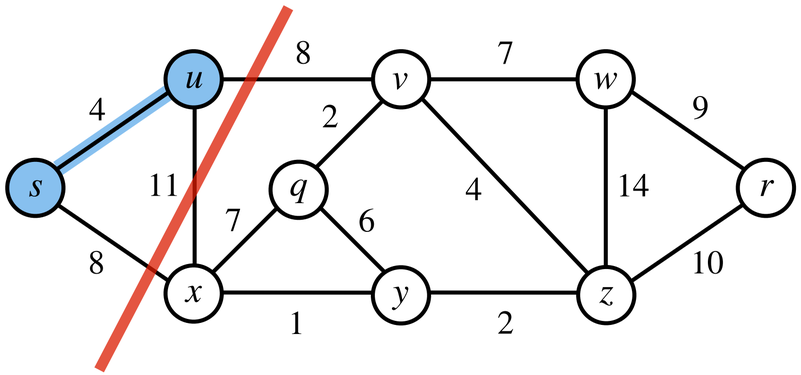

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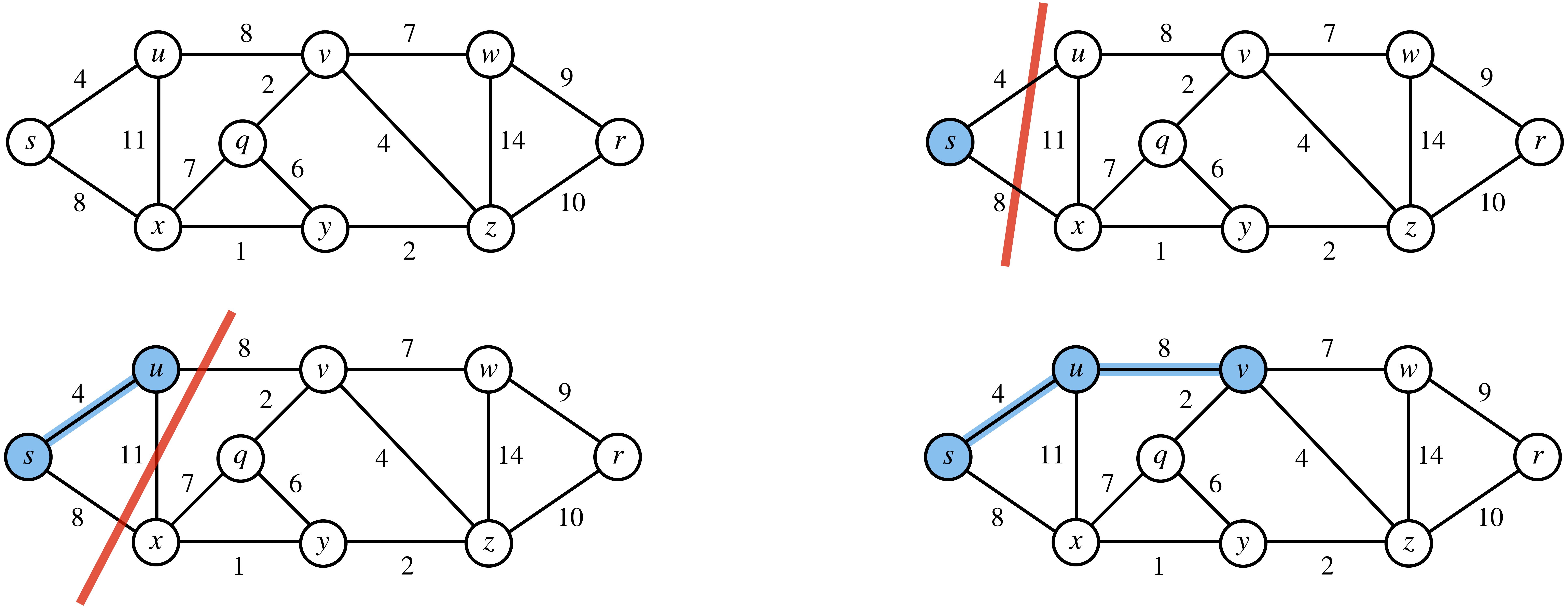

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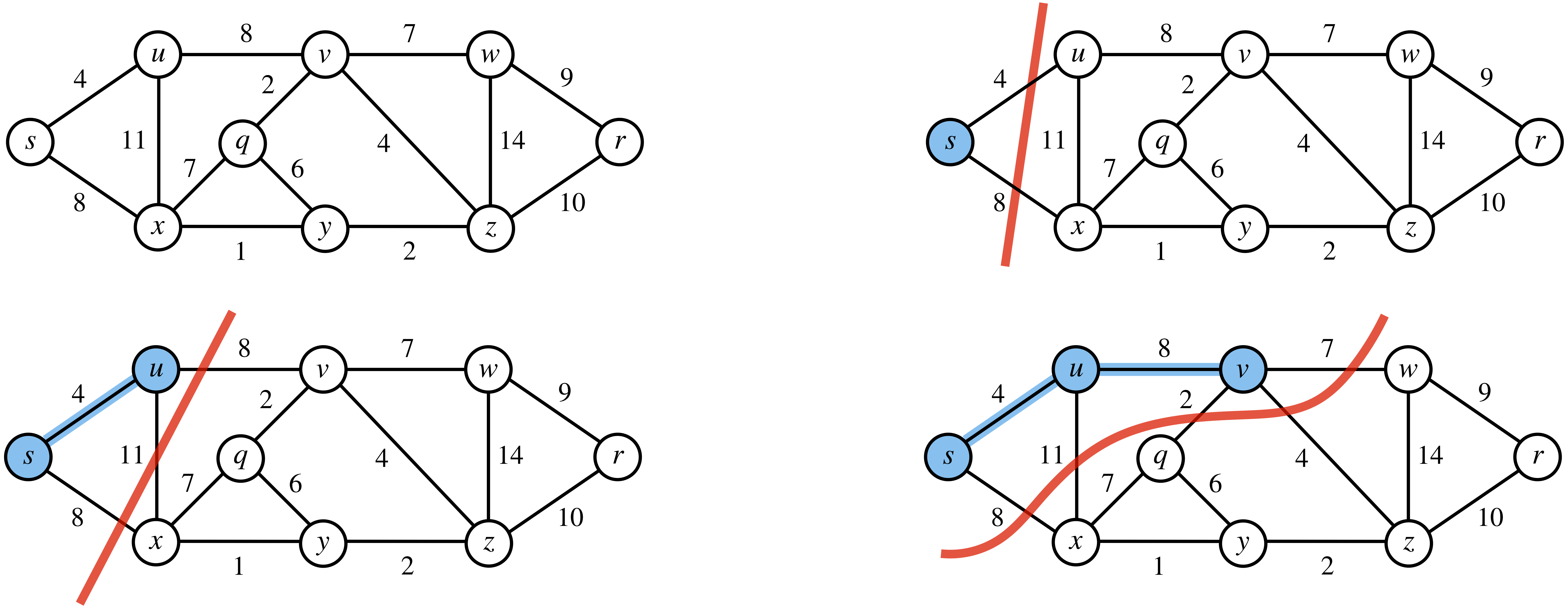

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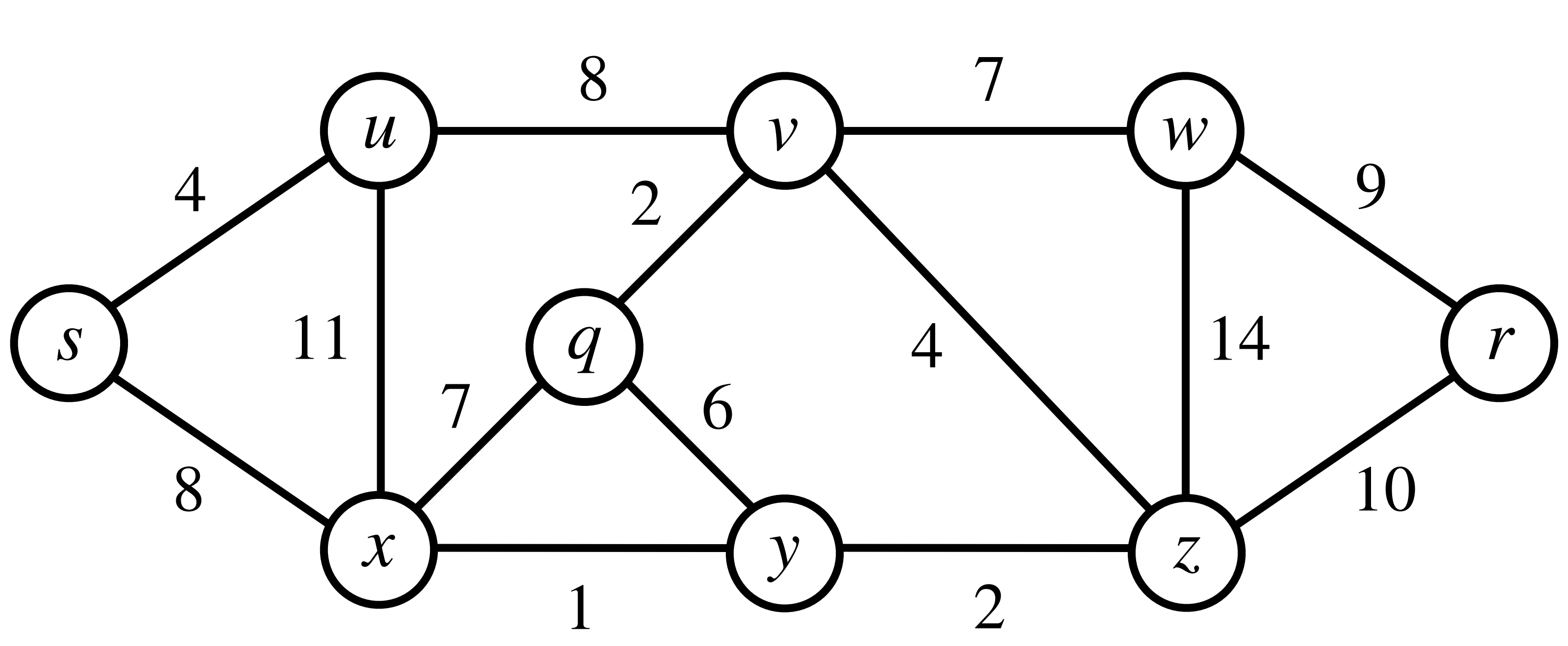

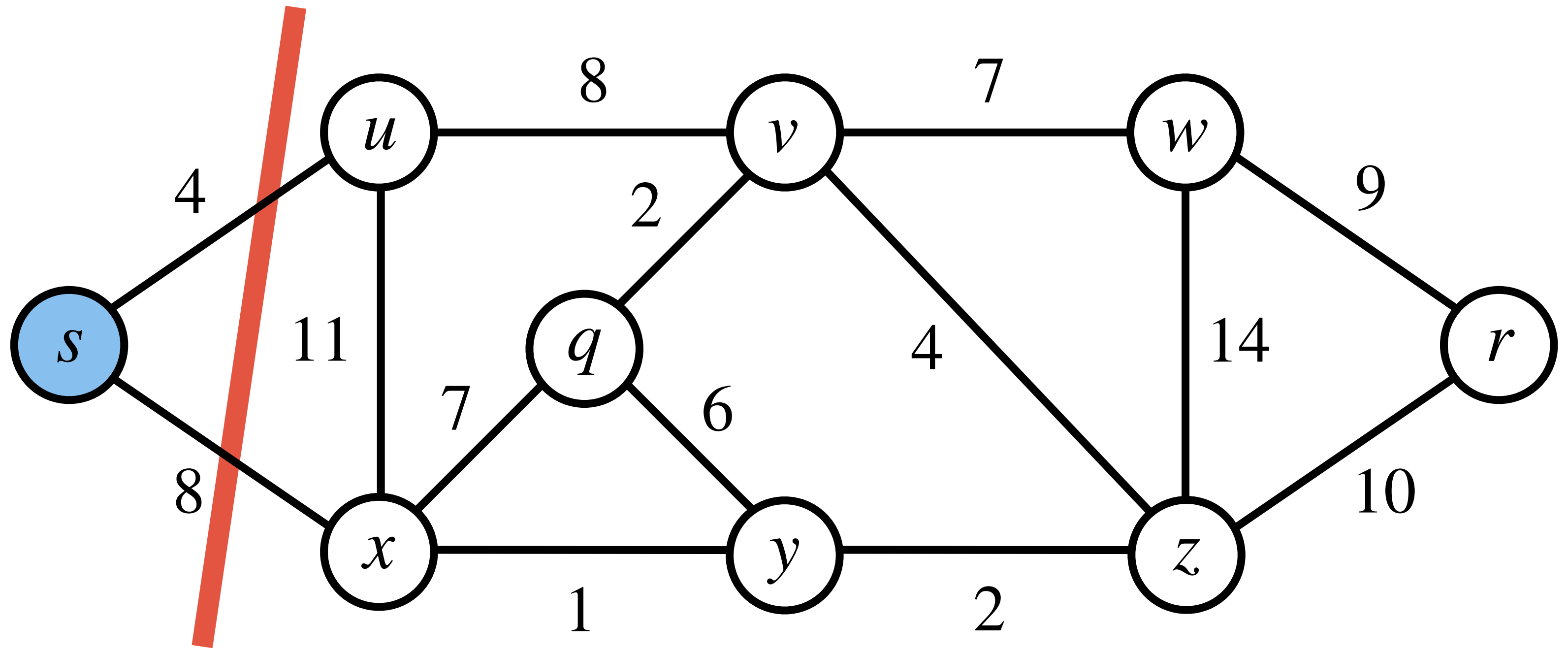

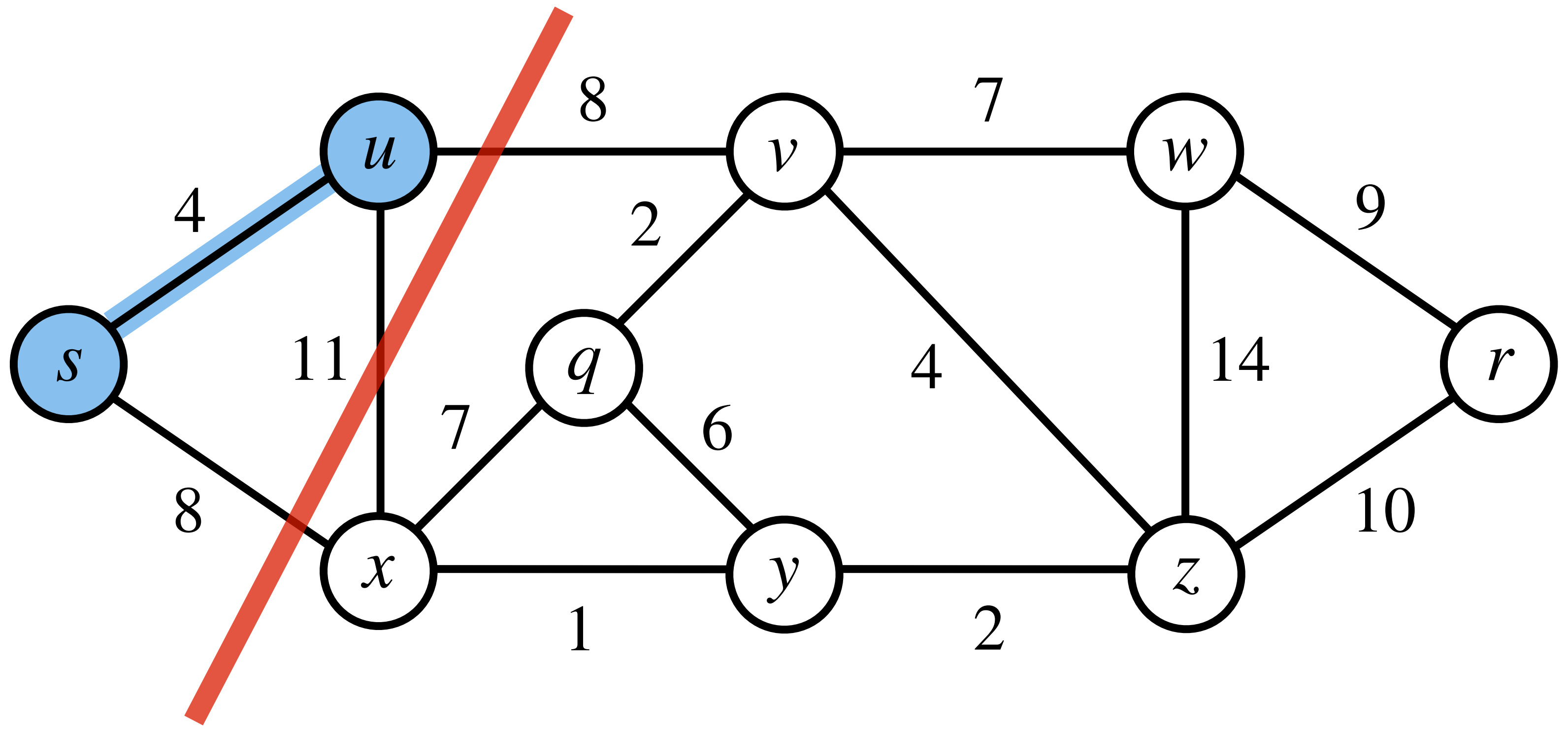

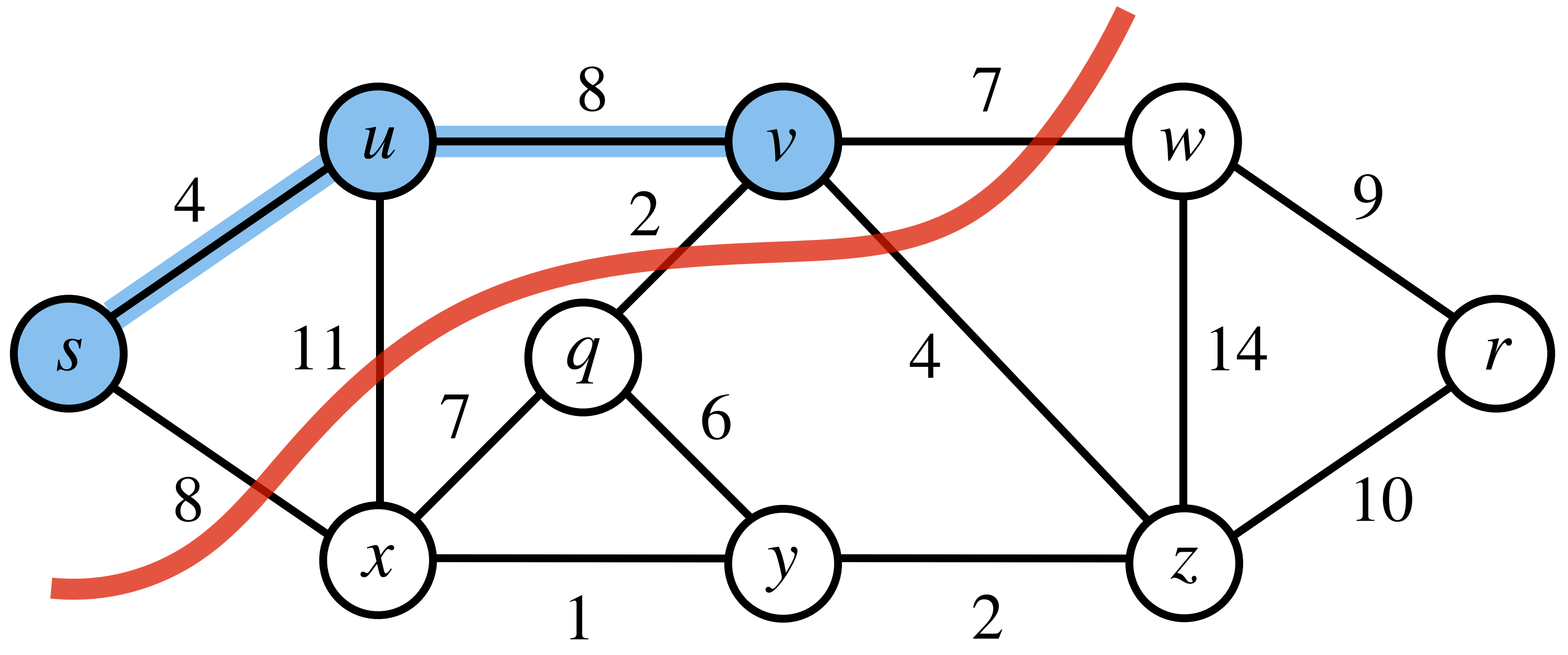

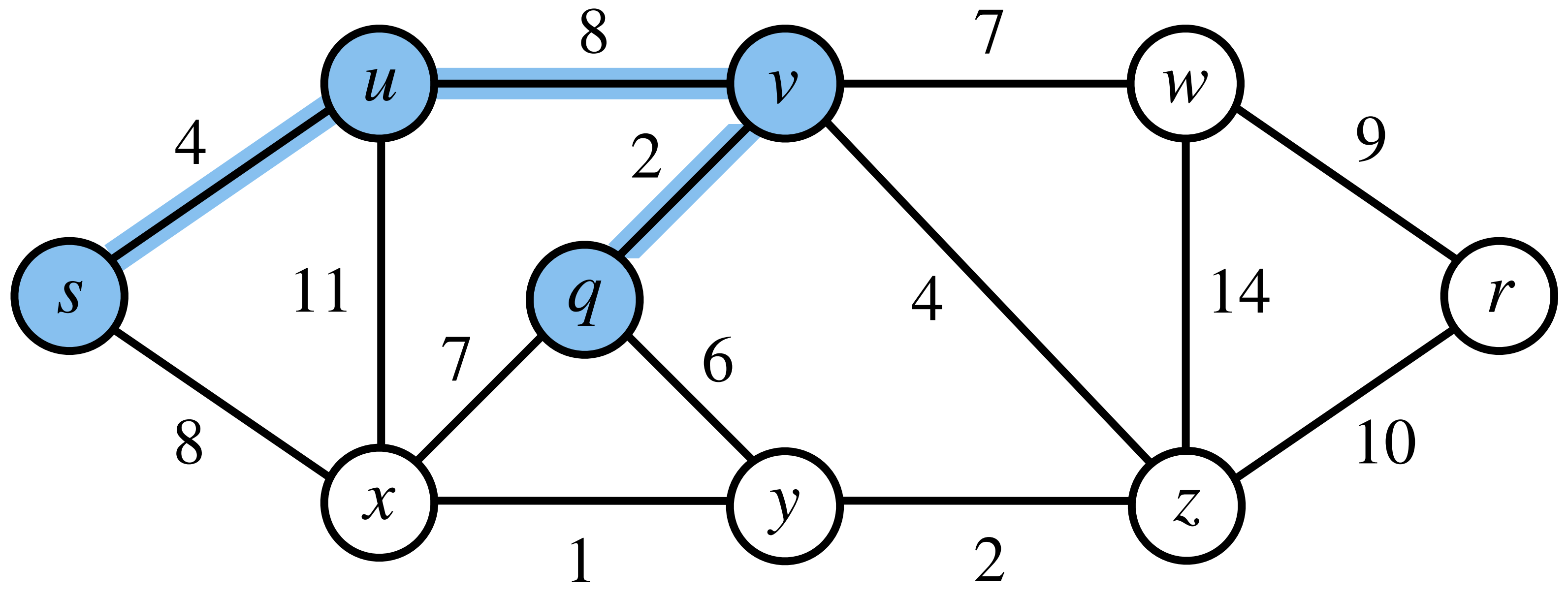

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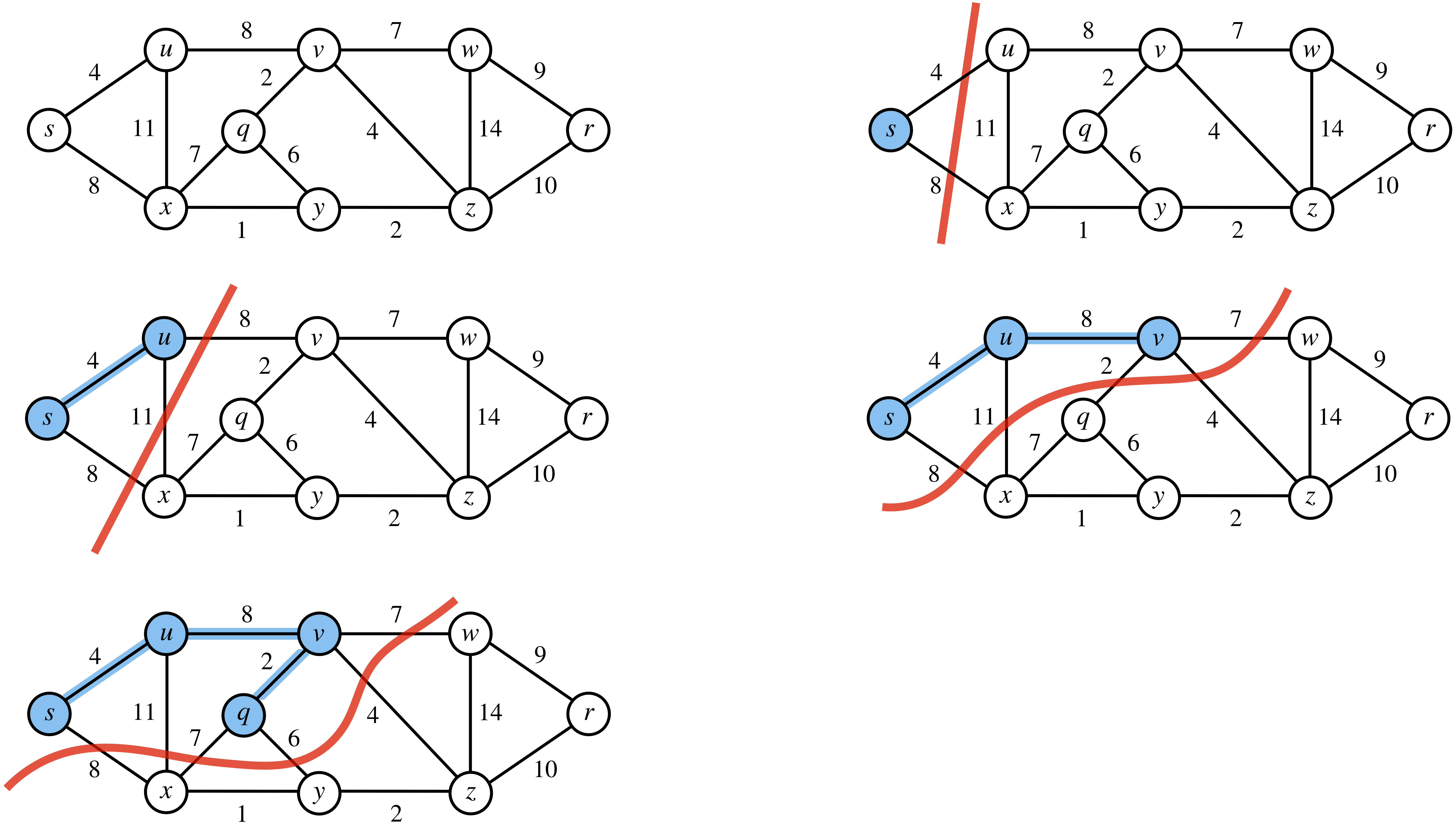

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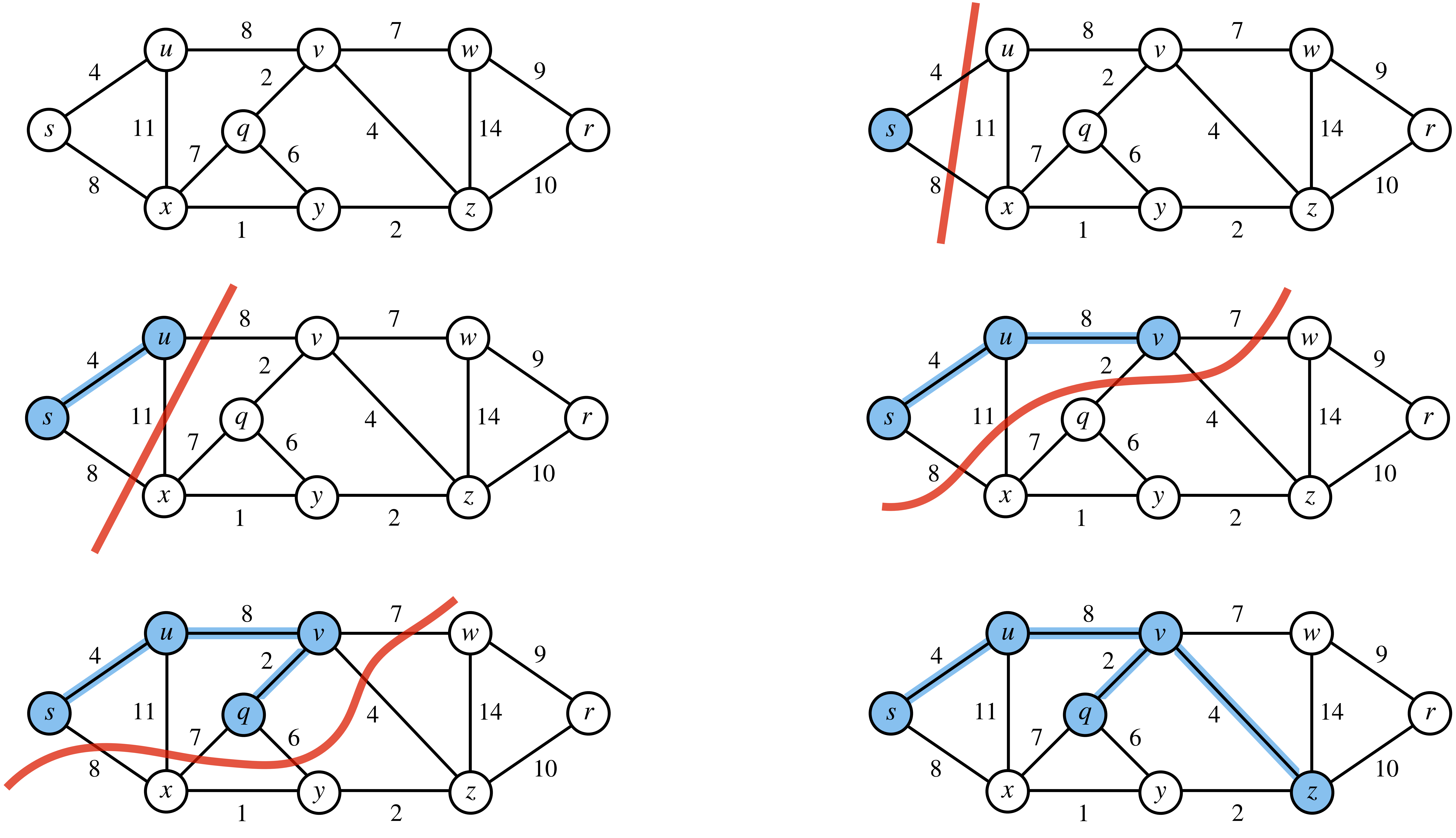

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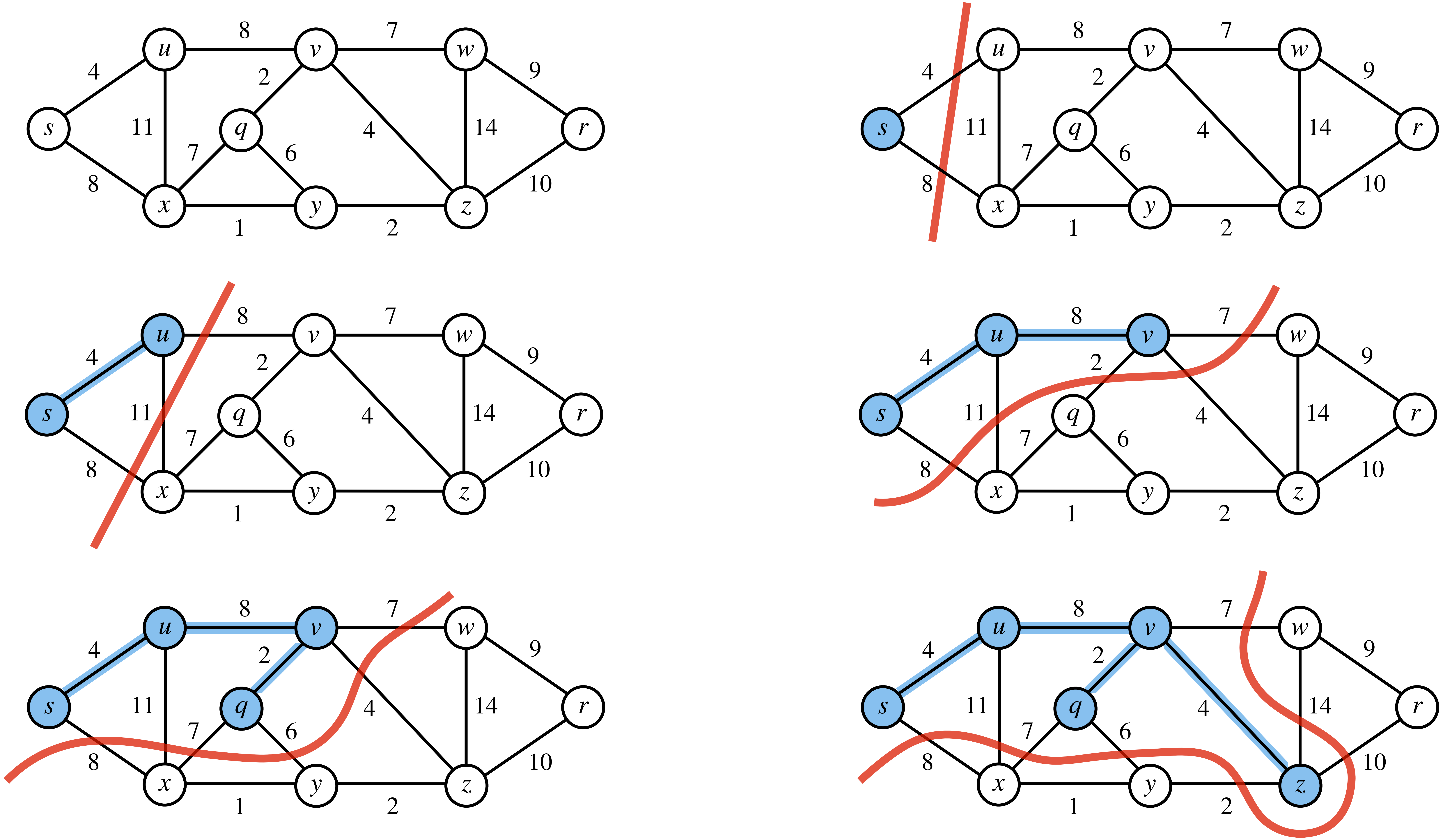

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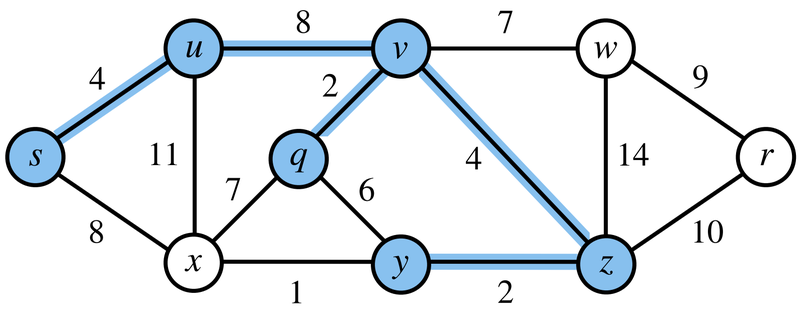

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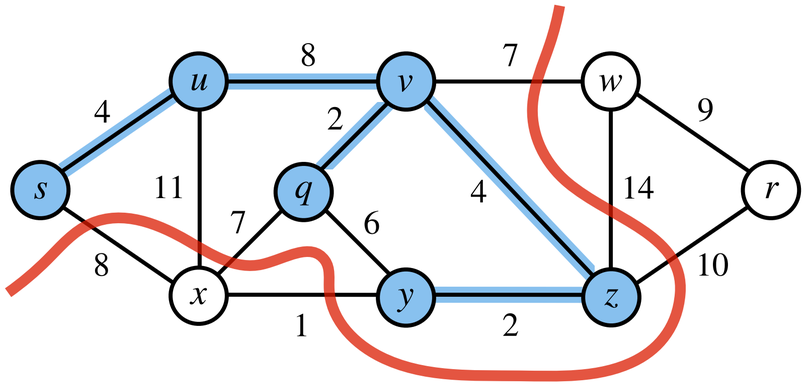

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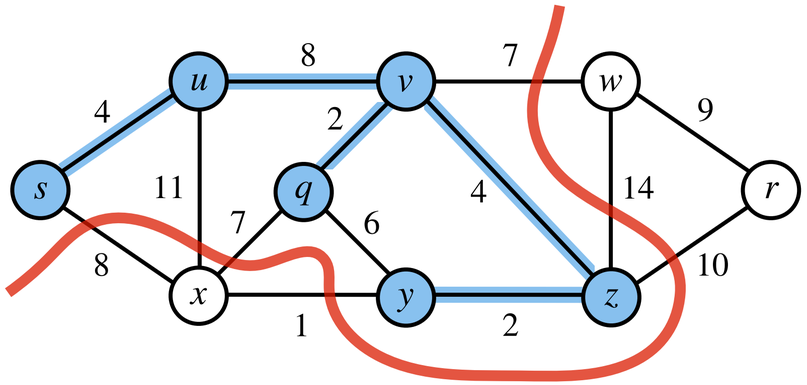

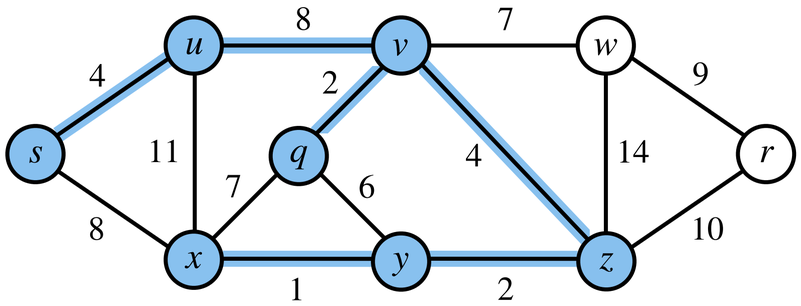

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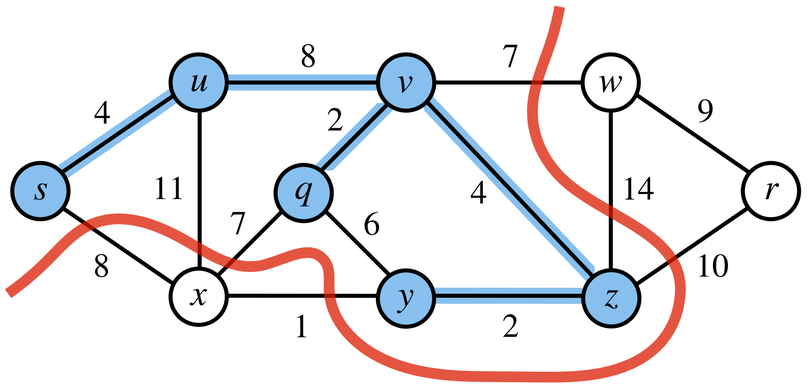

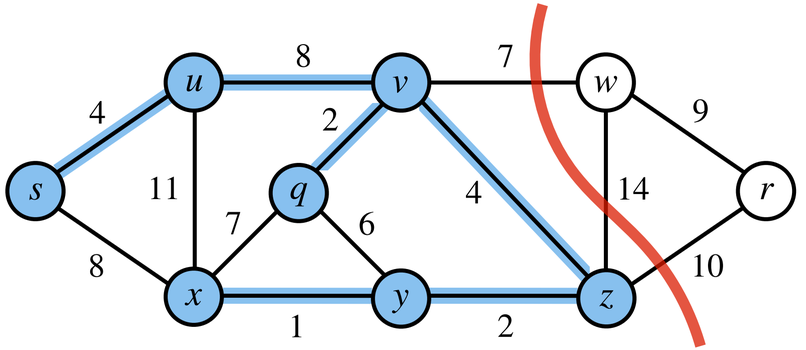

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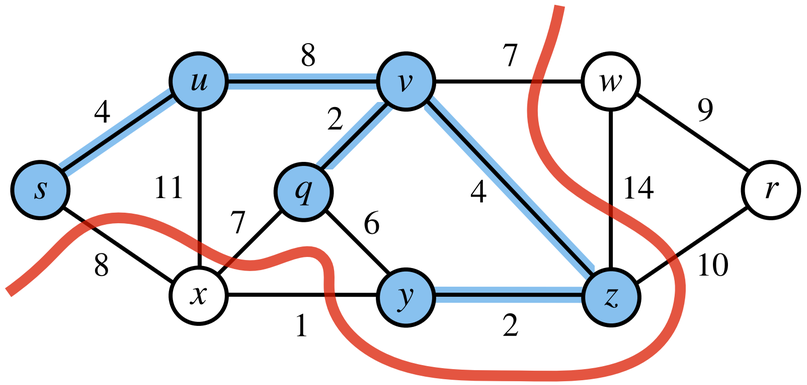
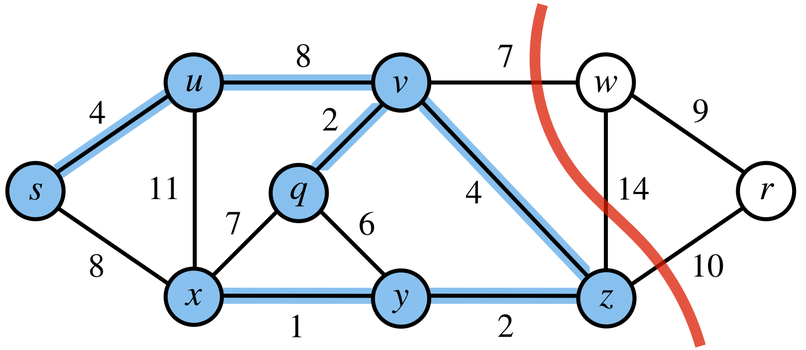
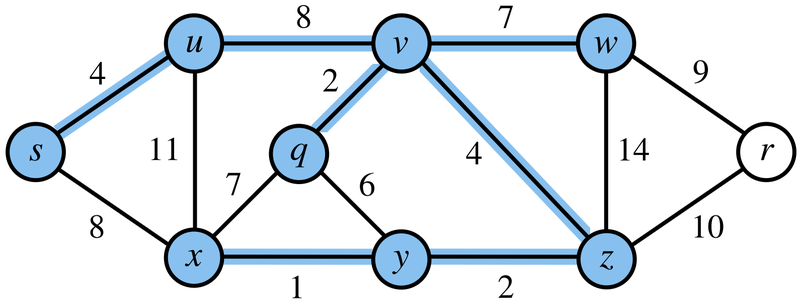
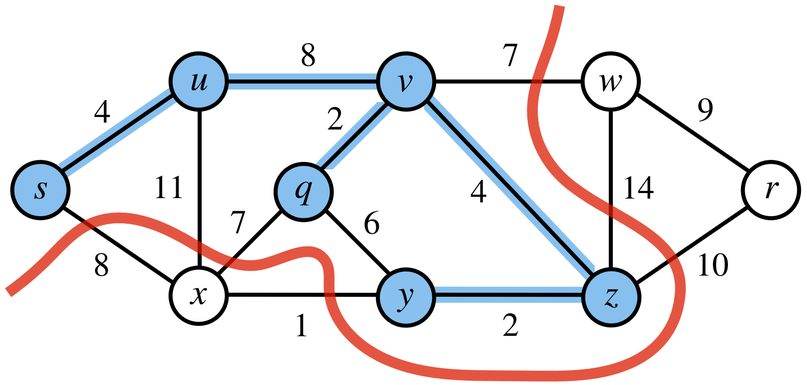
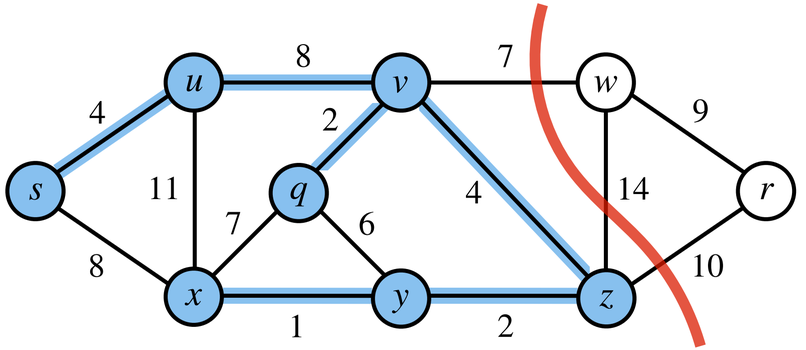


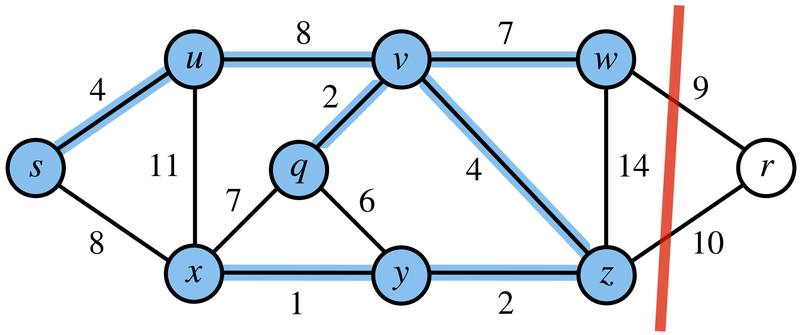




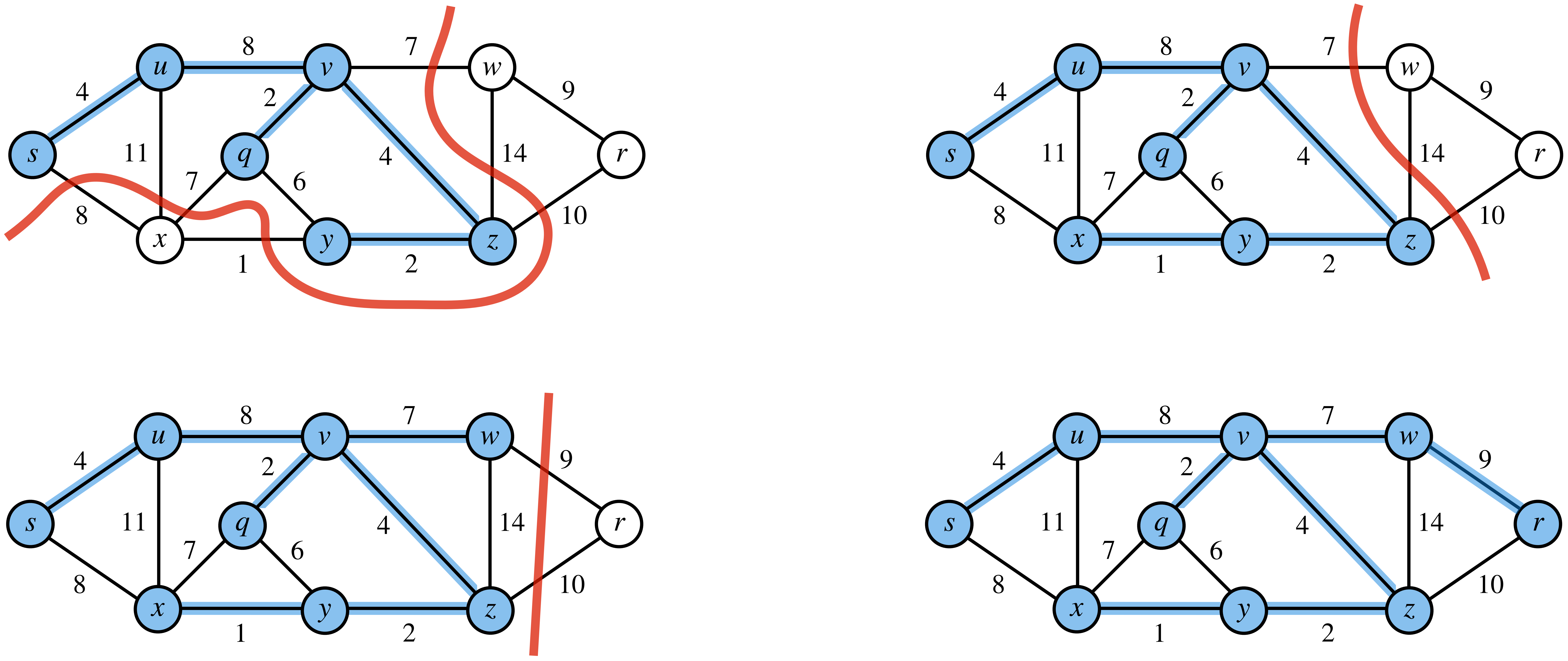
Prim's Algorithm: Demonstration







Prim's Algorithm: Demonstration



Prim’s Algorithm: Implementation

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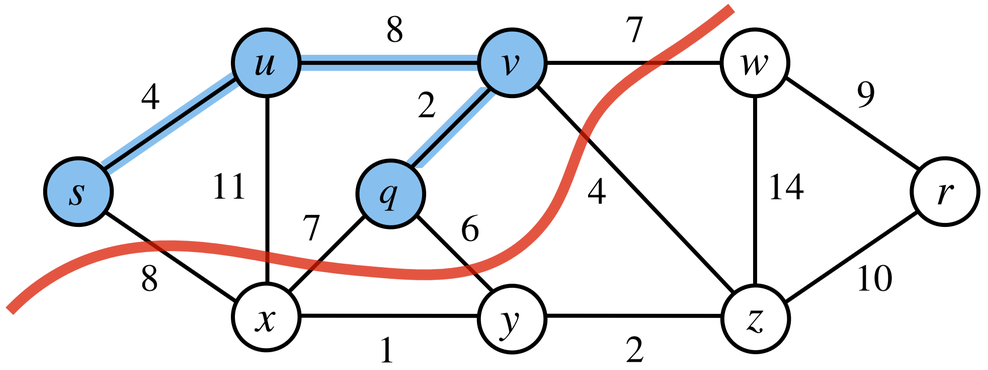
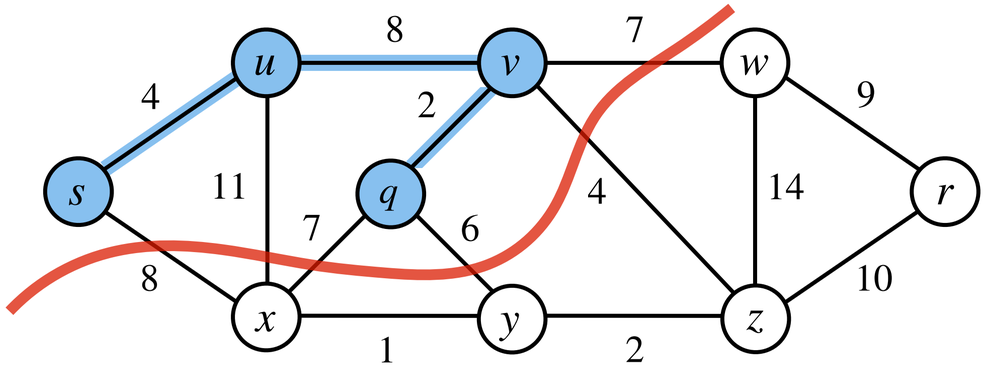
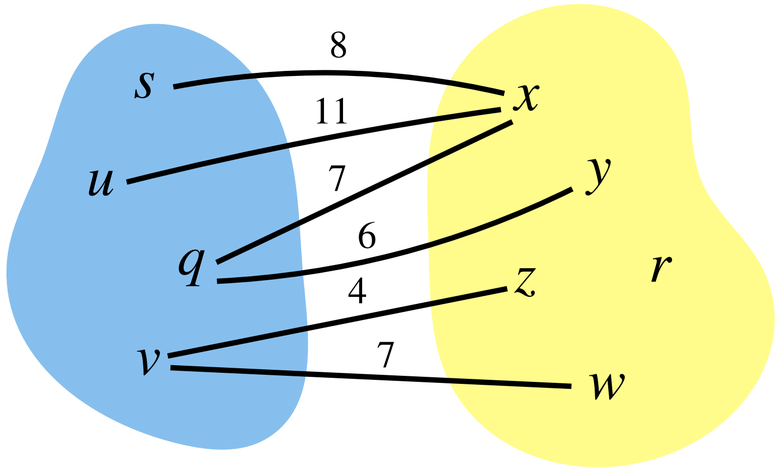
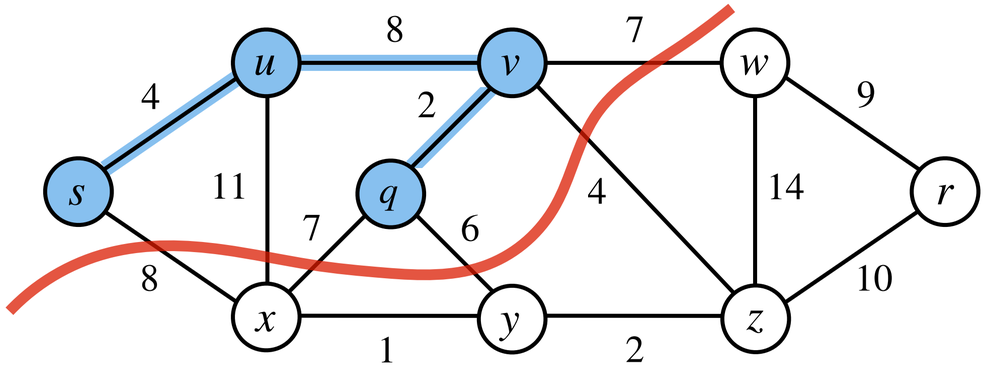


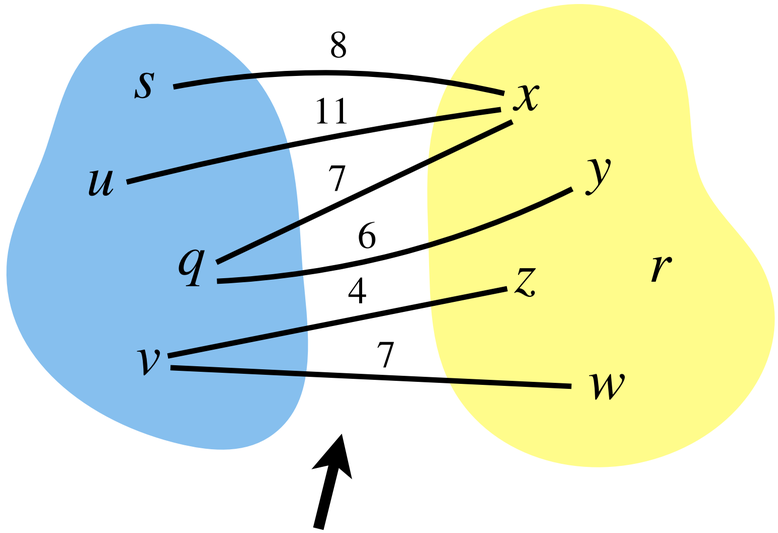
Prim's Algorithm: Implementation





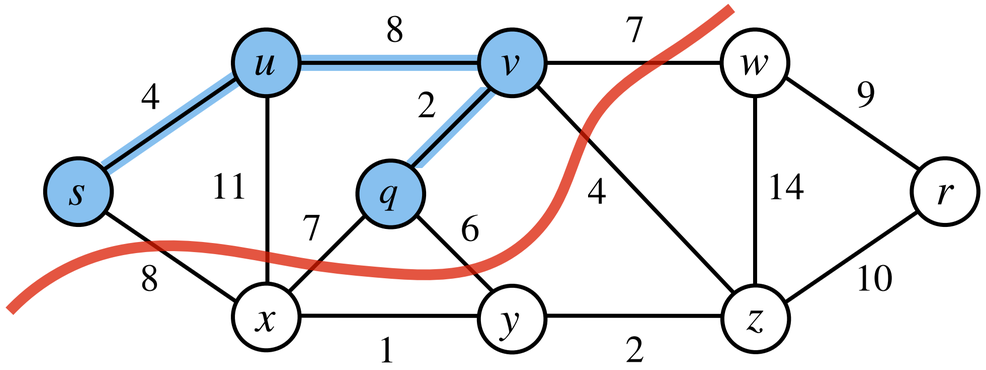
Prim's Algorithm: Implementation

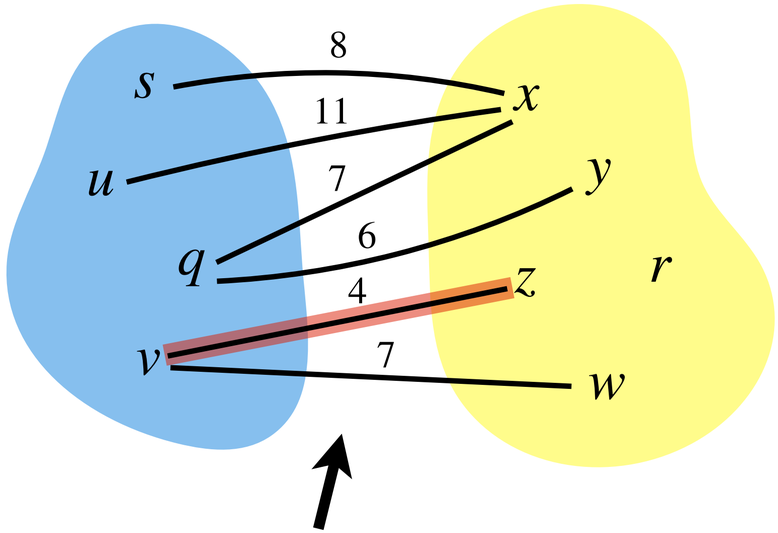




Find a least weight edge straightforwardly.

Prim's Algorithm: Implementation





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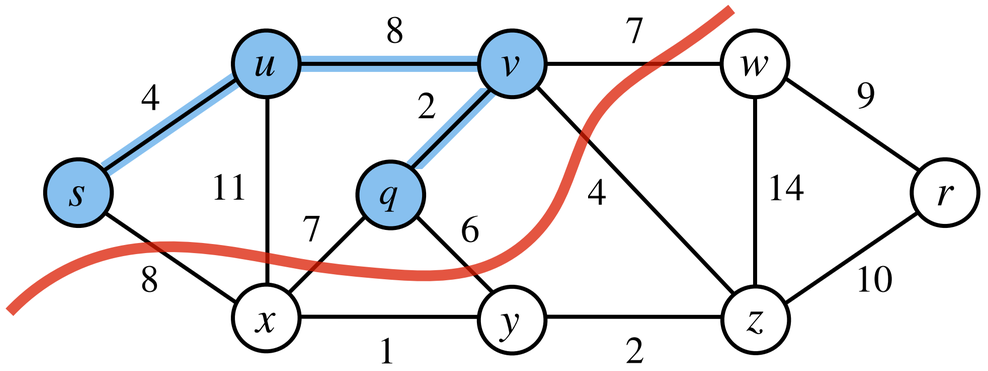

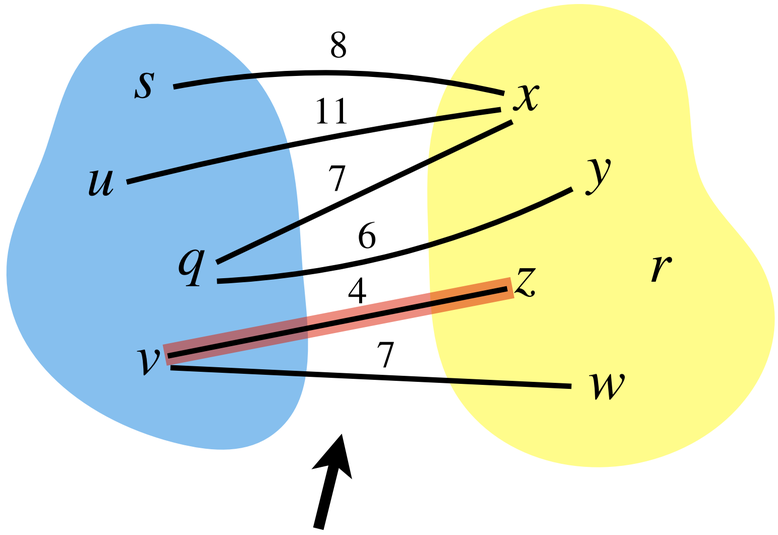

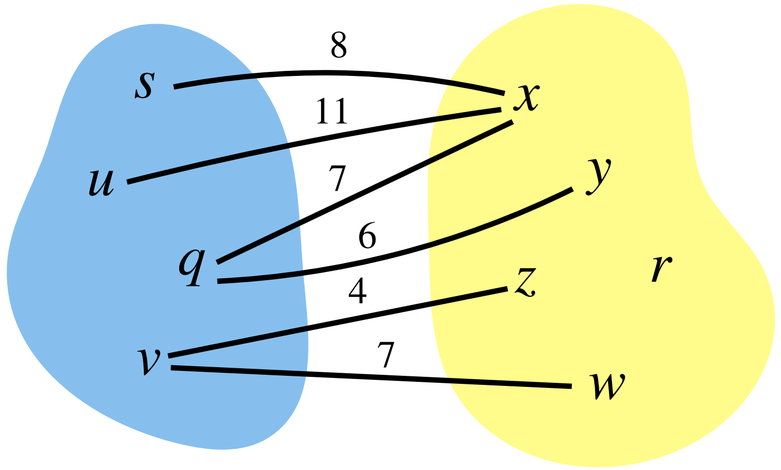

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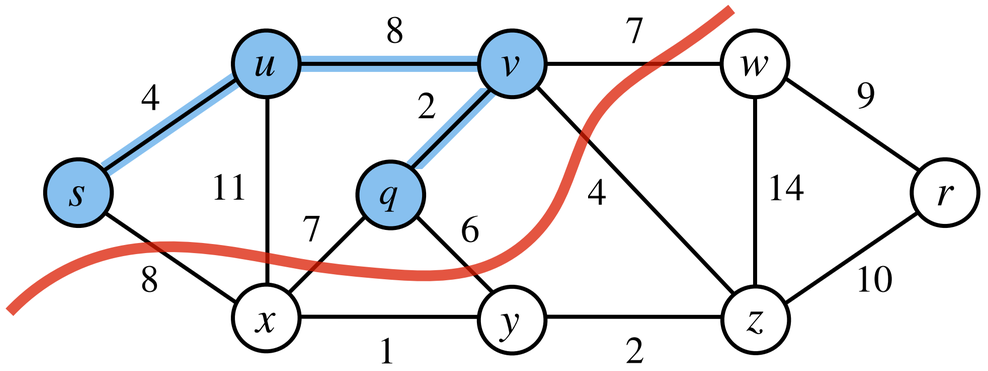
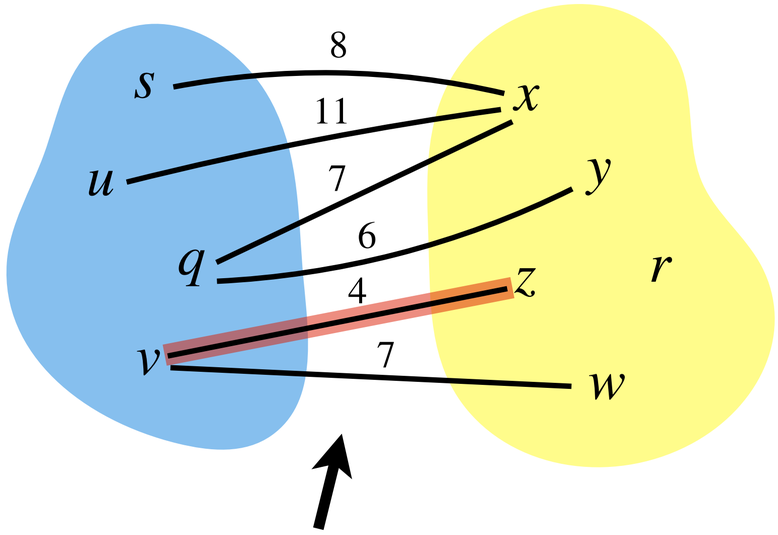
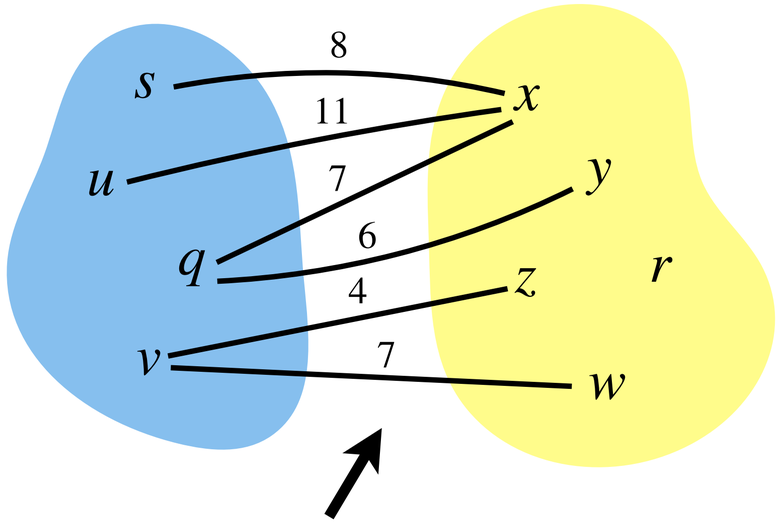


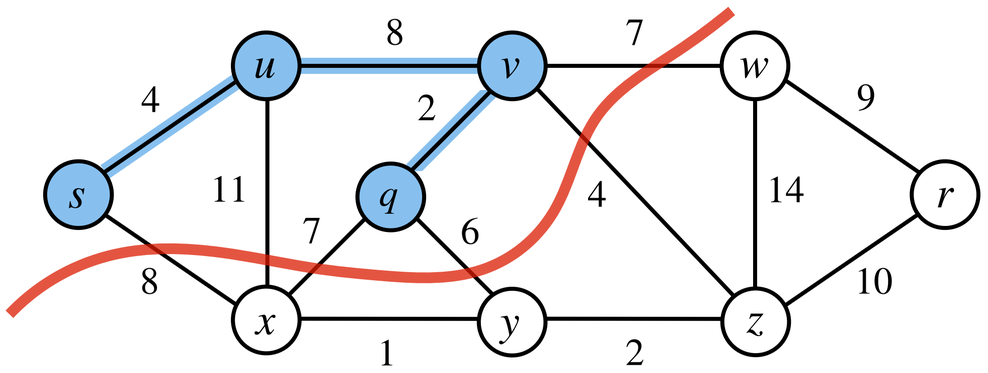


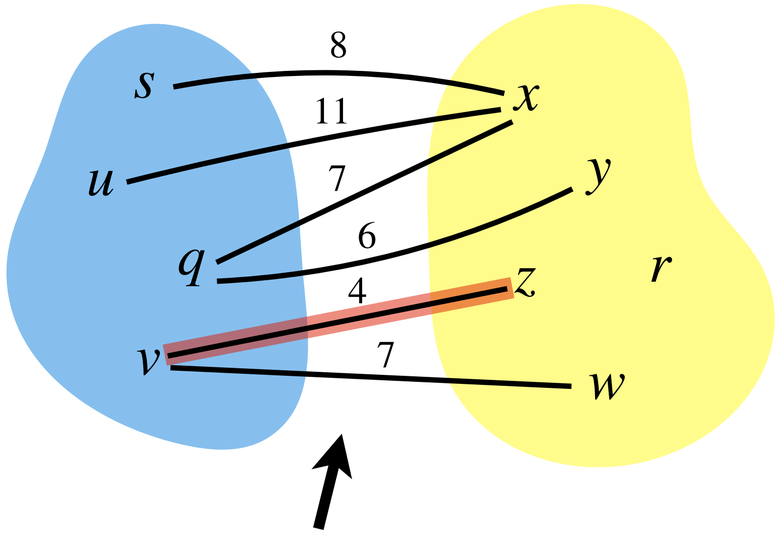
Find a least weight edge straightforwardly.



For every vertex not in tree, maintain a *key* that is equal to the weight of a least weight edge from a tree vertex to it.

Prim's Algorithm: Implementation





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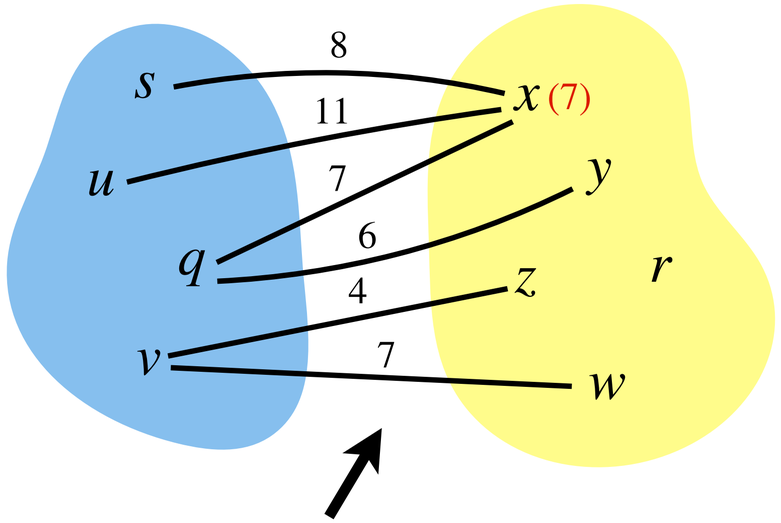

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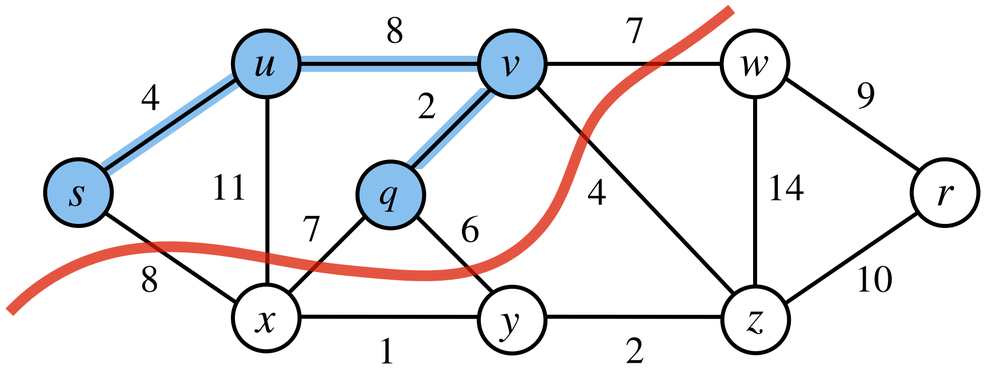

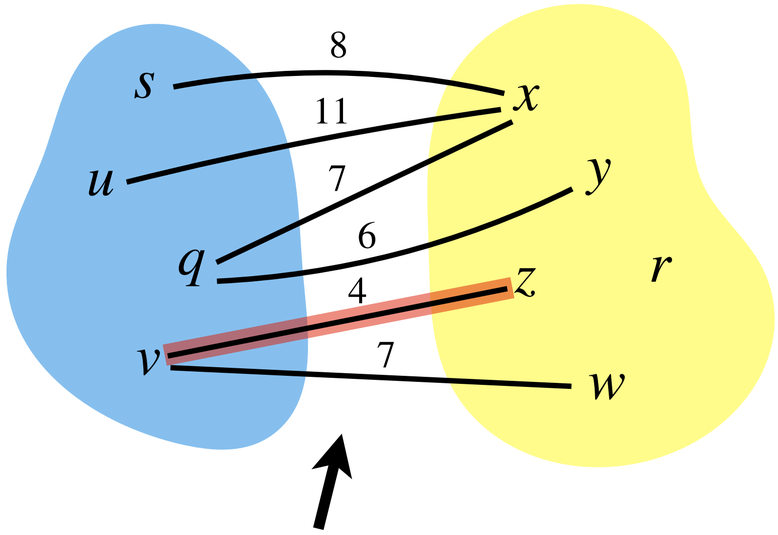

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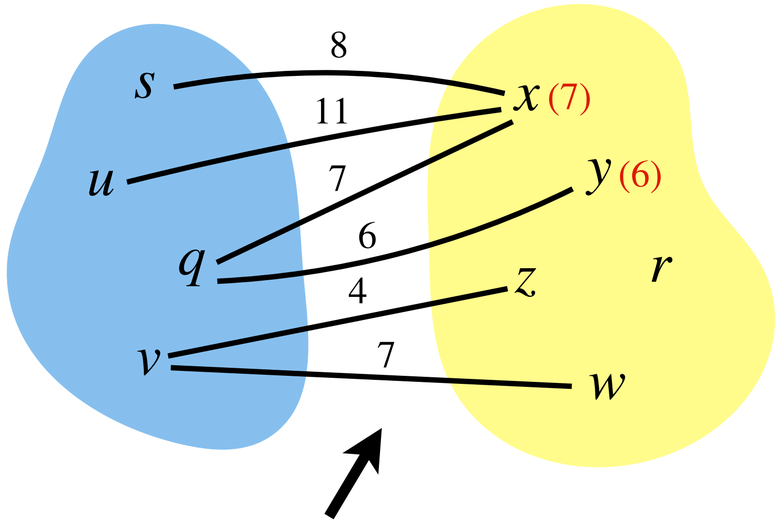

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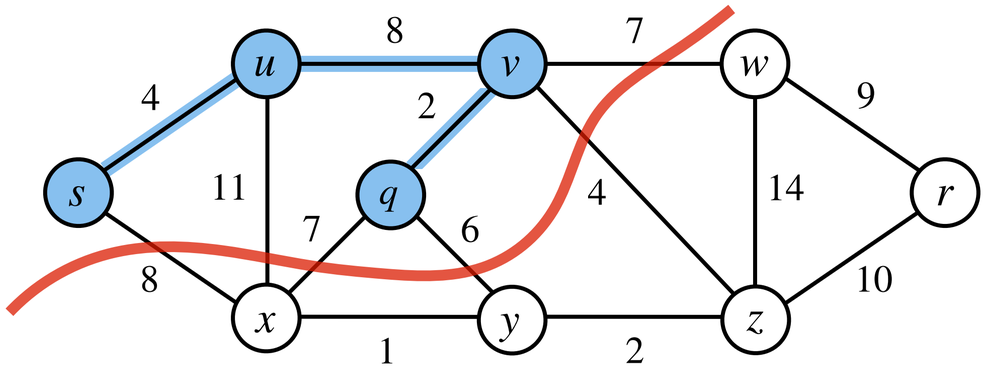
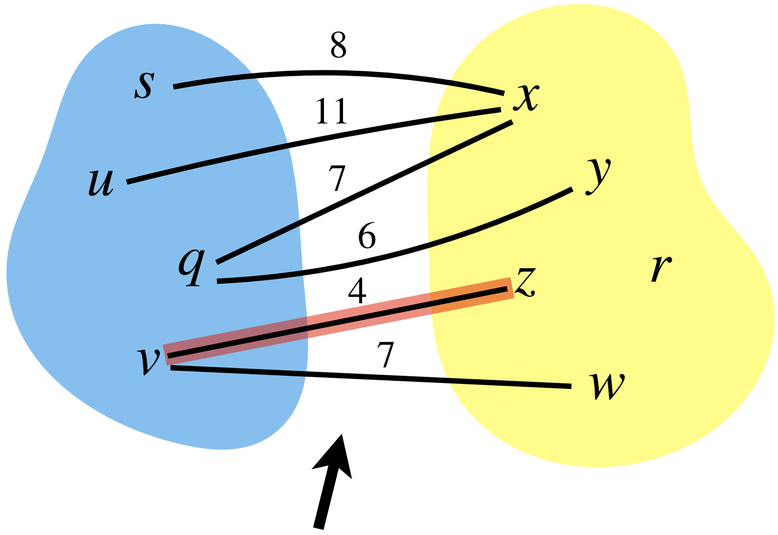
Prim's Algorithm: Implementation





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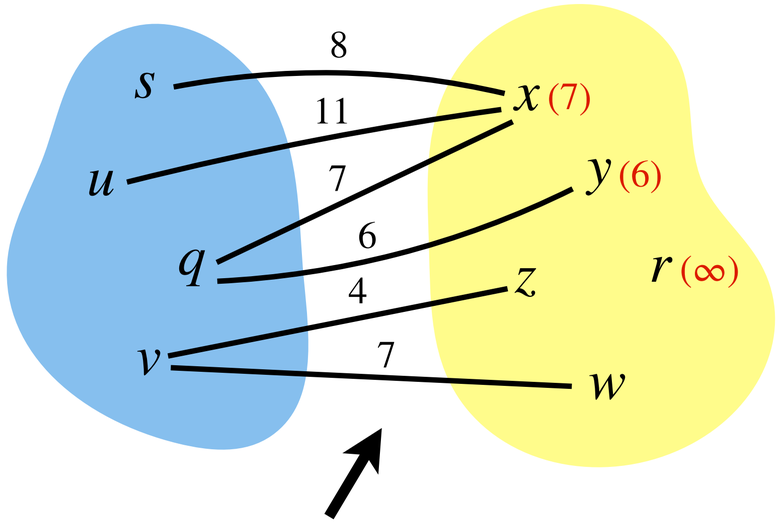

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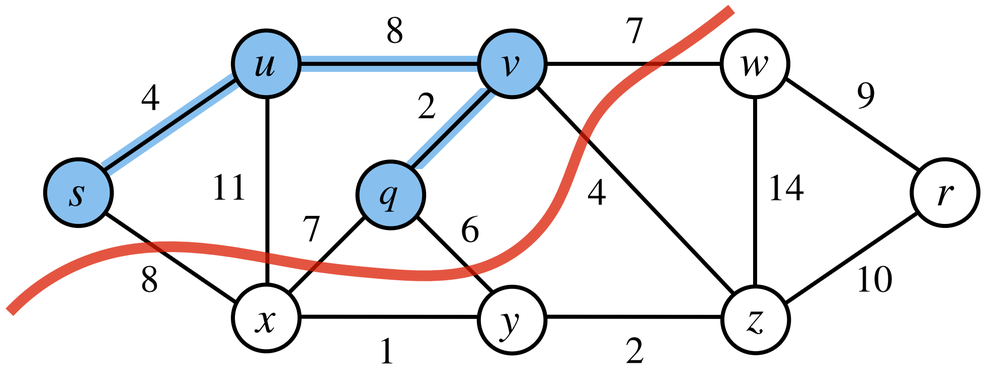

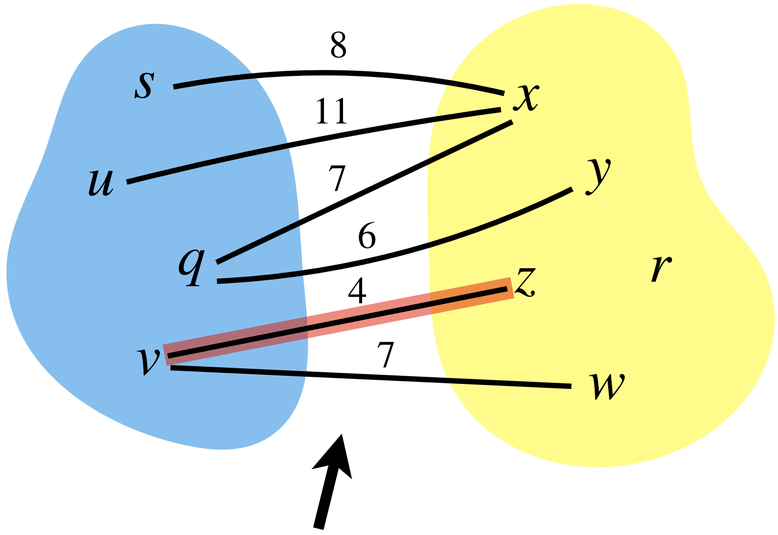

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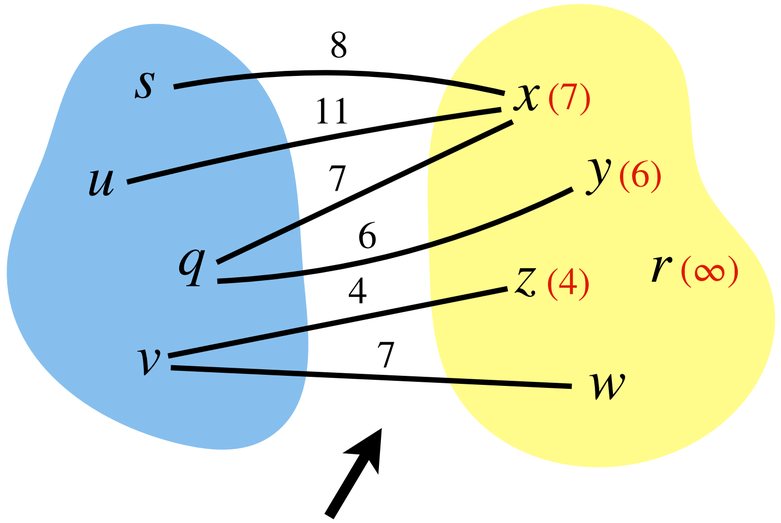

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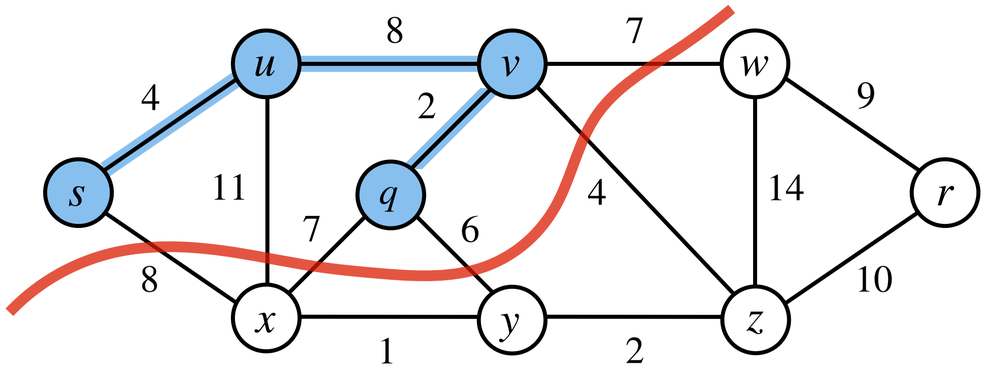

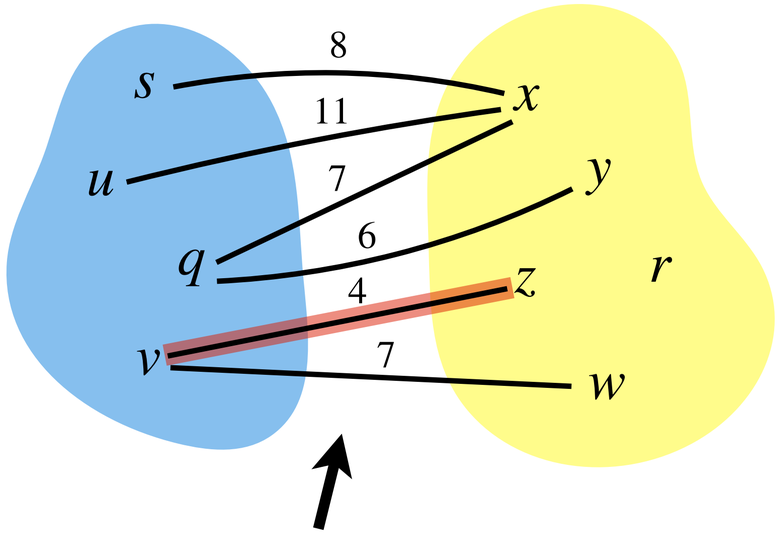

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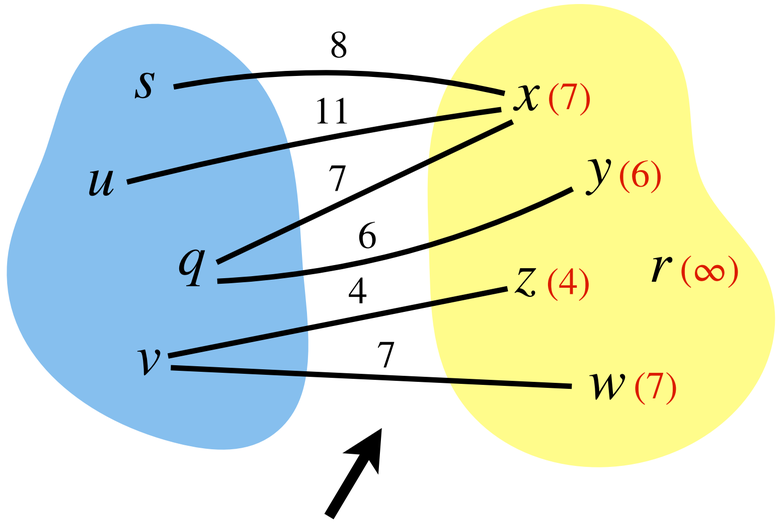

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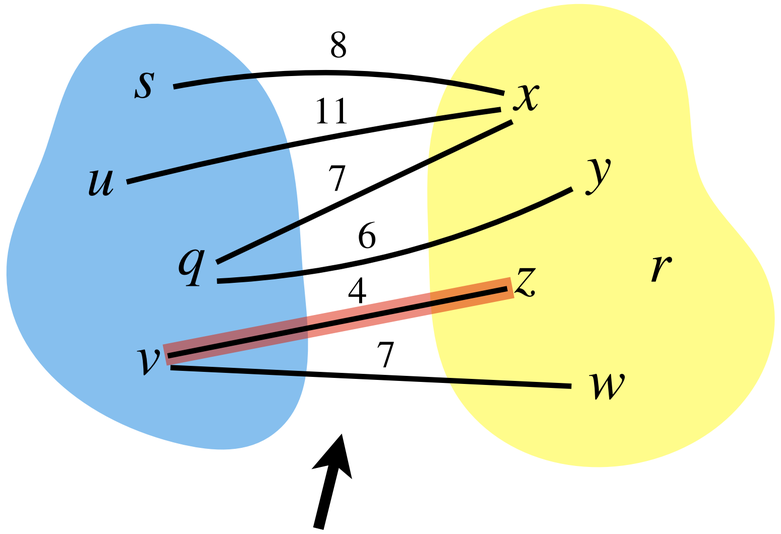

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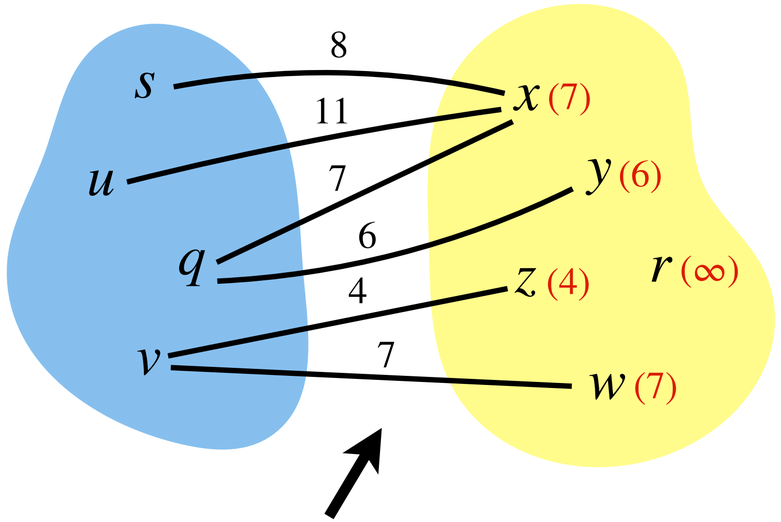

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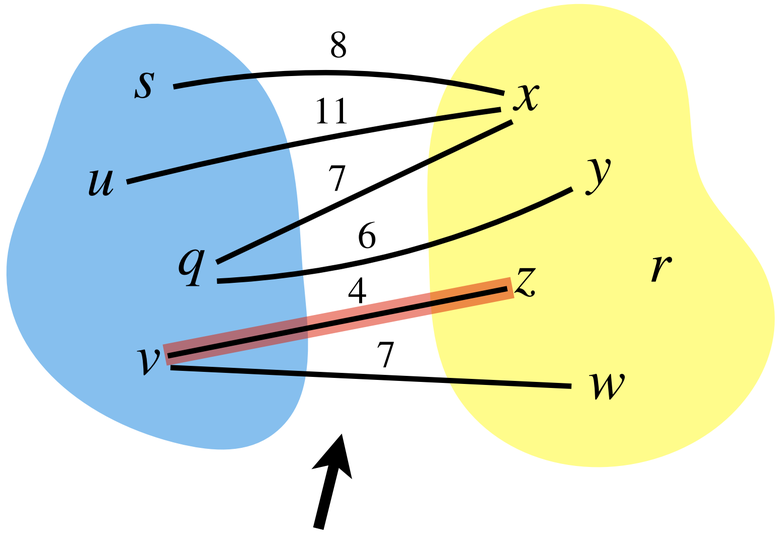

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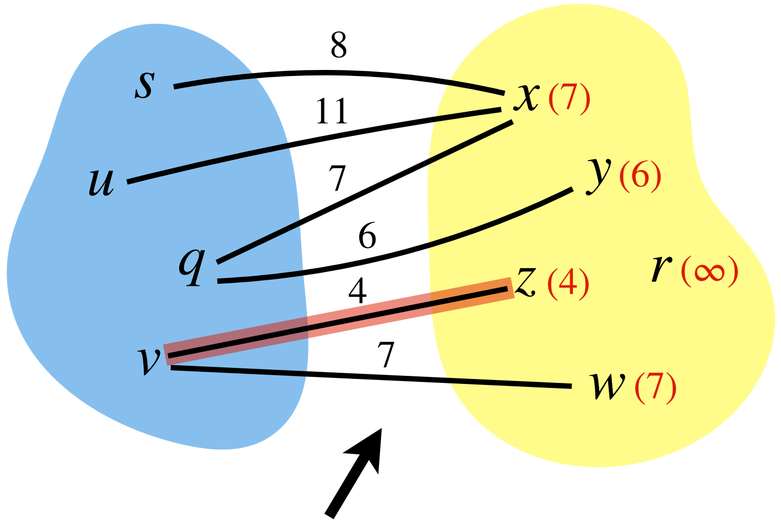

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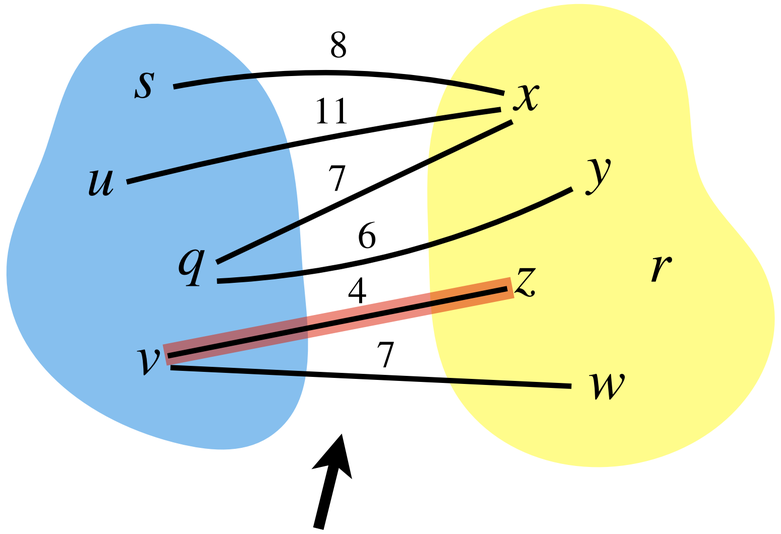

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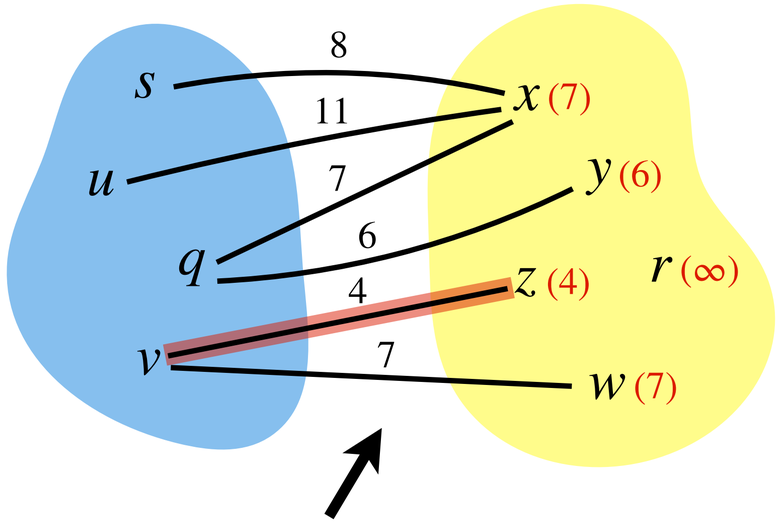
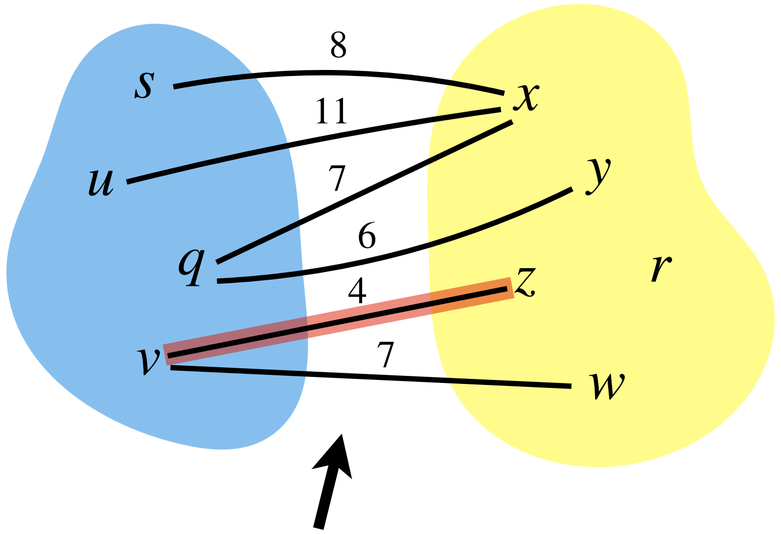


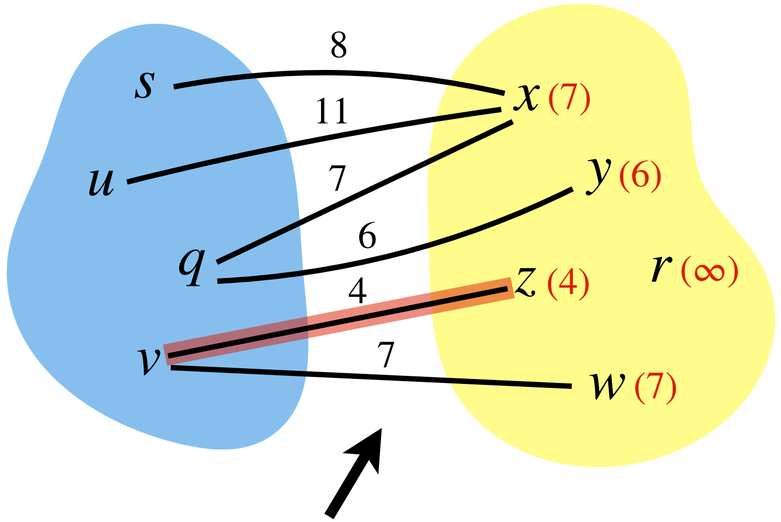
For every vertex not in tree, maintain a *key* that is equal to the weight of a least weight edge from a tree vertex to it.

Observation: Finding a least weight edge in the cut-set is the same as finding a non-tree

Prim's Algorithm: Implementation



Find a least weight edge straightforwardly.

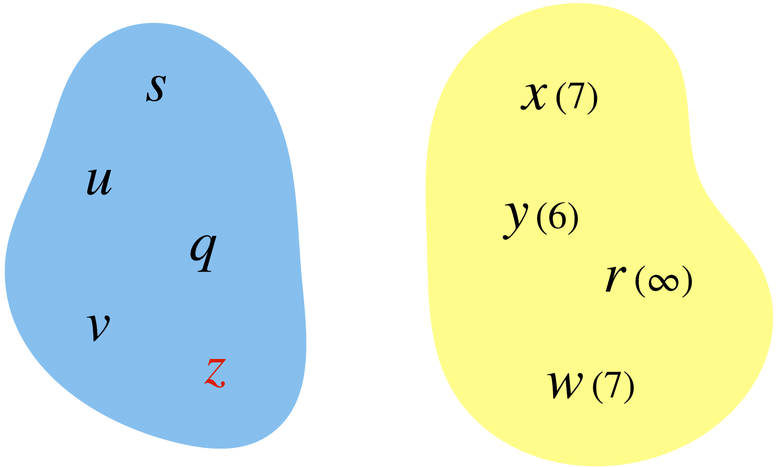


For every vertex not in tree, maintain a *key* that is equal to the weight of a least weight edge from a tree vertex to it.

Observation: Finding a least weight edge in the cut-set is the same as finding a non-tree vertex with minimum *key* and the edge which connects it to tree.

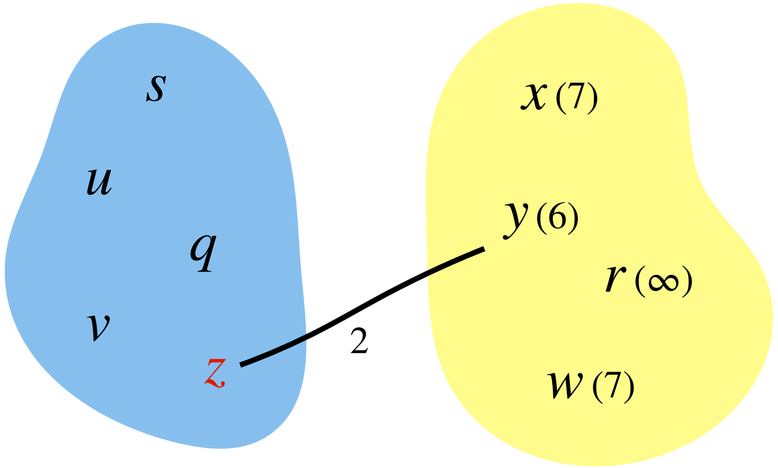
Prim's Algorithm: Implementation

Updating *keys*:



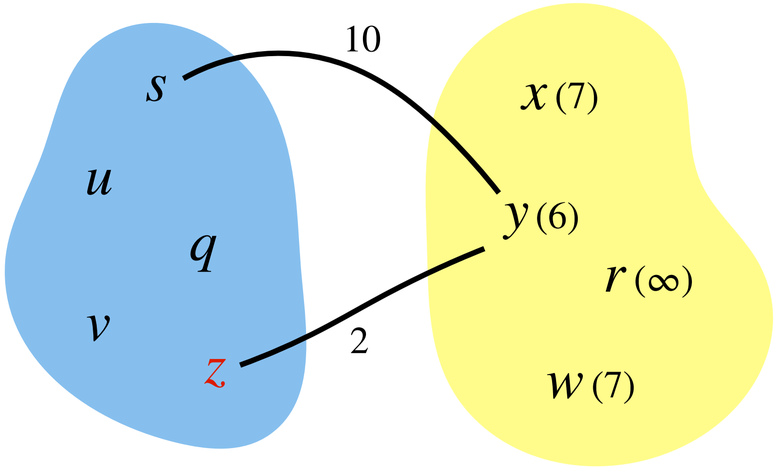
Prim's Algorithm: Implementation

Updating *keys*:



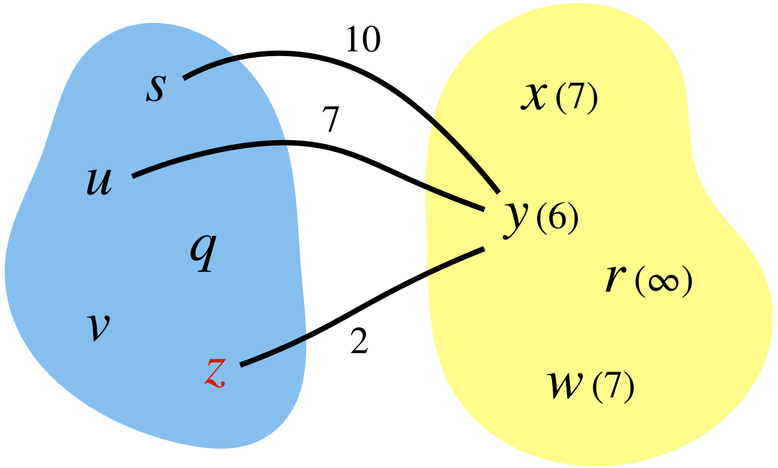
Prim's Algorithm: Implementation

Updating *keys*:



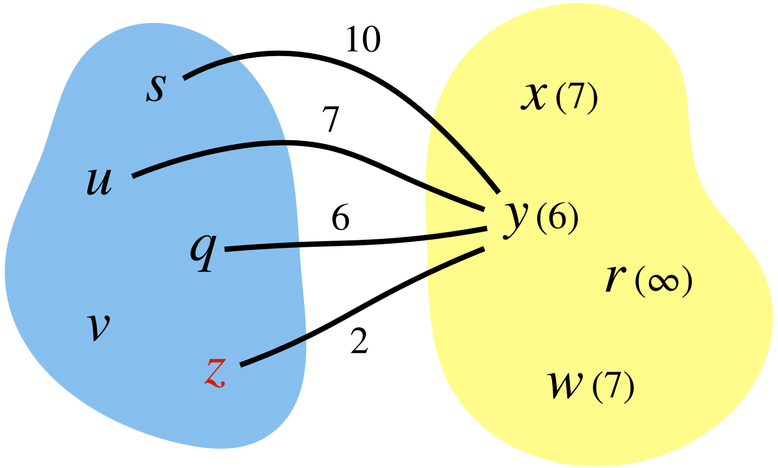
Prim's Algorithm: Implementation

Updating *keys*:



Prim's Algorithm: Implementation

Updating *keys*:



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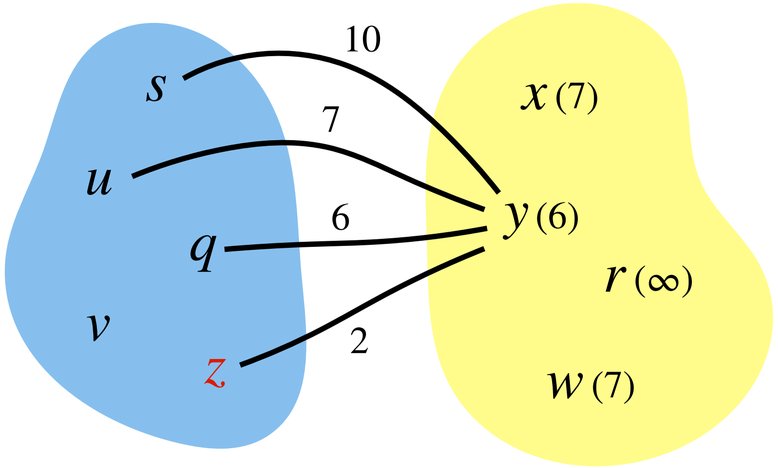
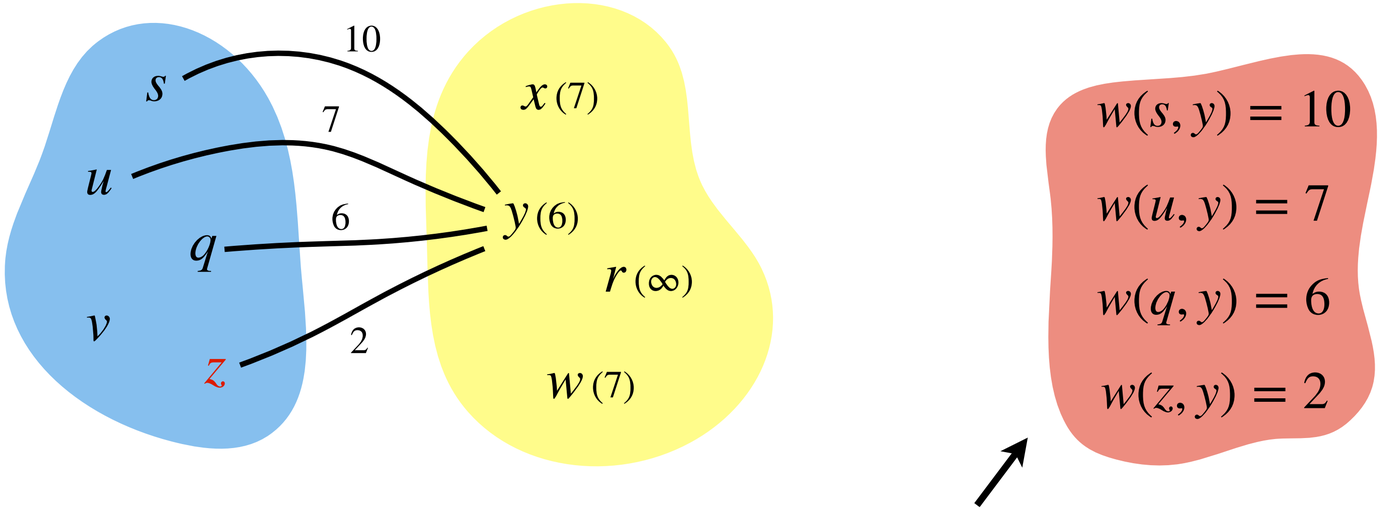
Updating *keys*:



key[*y*] will be minimum of these values.

Prim's Algorithm: Implementation

Updating *keys*:



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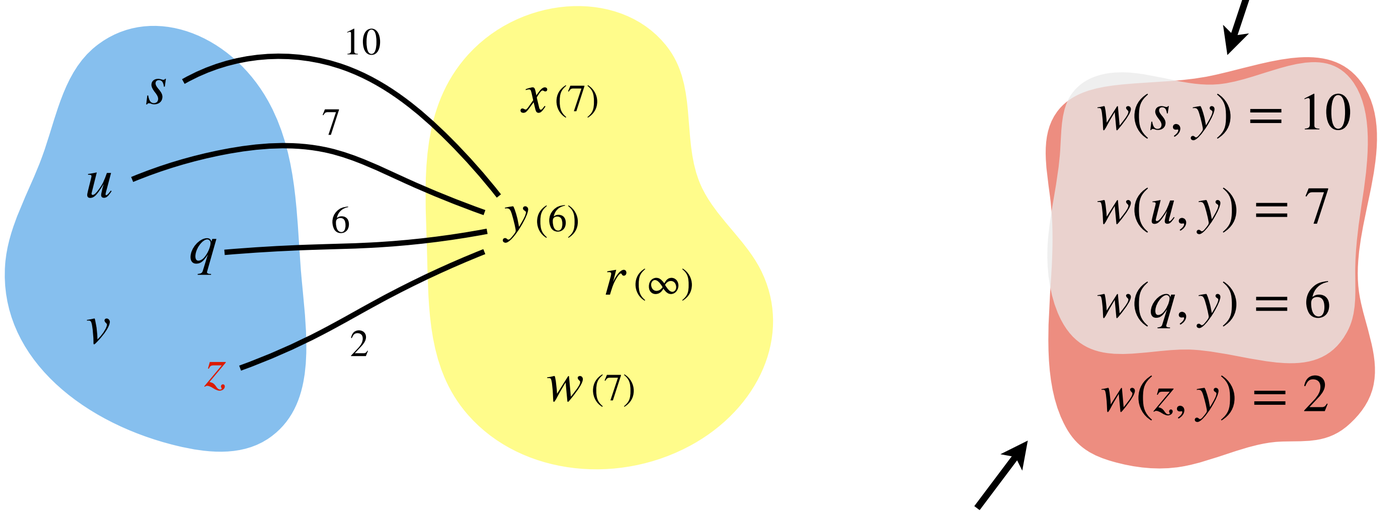

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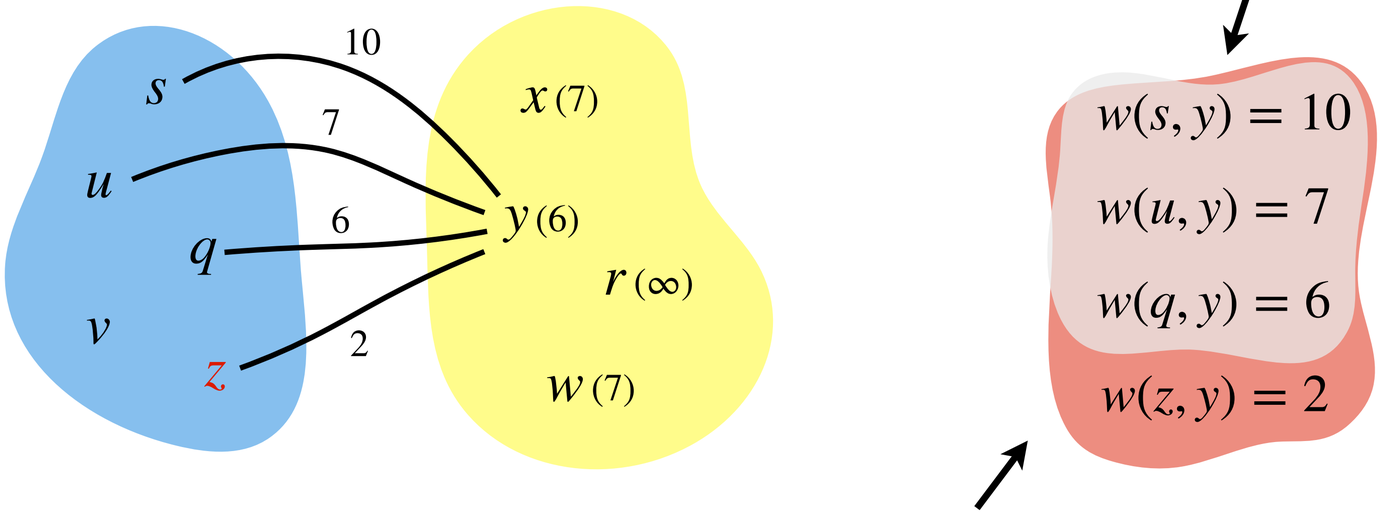

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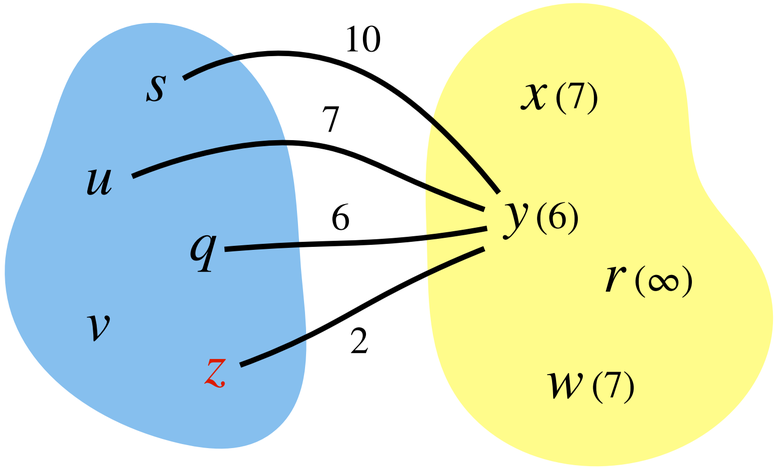

Minimum of these is old $key[y]$

$w(s, y) = 10$

$w(u, y) = 7$

$w(q, y) = 6$

$w(z, y) = 2$

$key[y]$ will be minimum of these values.

Observation: If u is just added to tree, then $\forall v$ not in tree, such that $\{u, v\}$ is an edge

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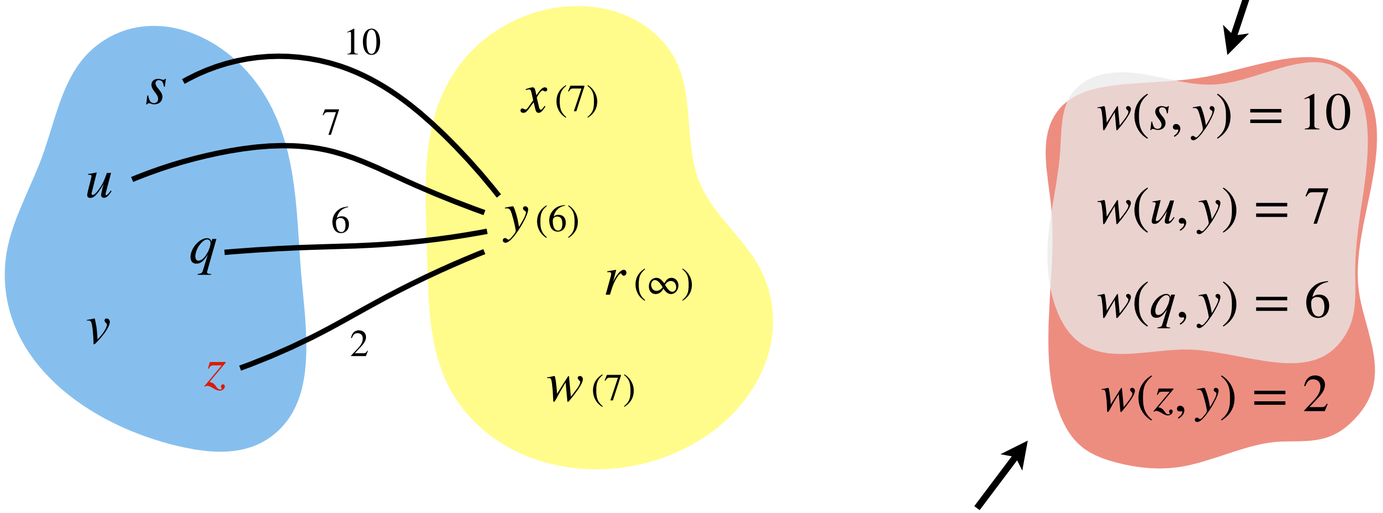

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$$key[v] = \mathbf{Min}(key[v], w(u, v))$$

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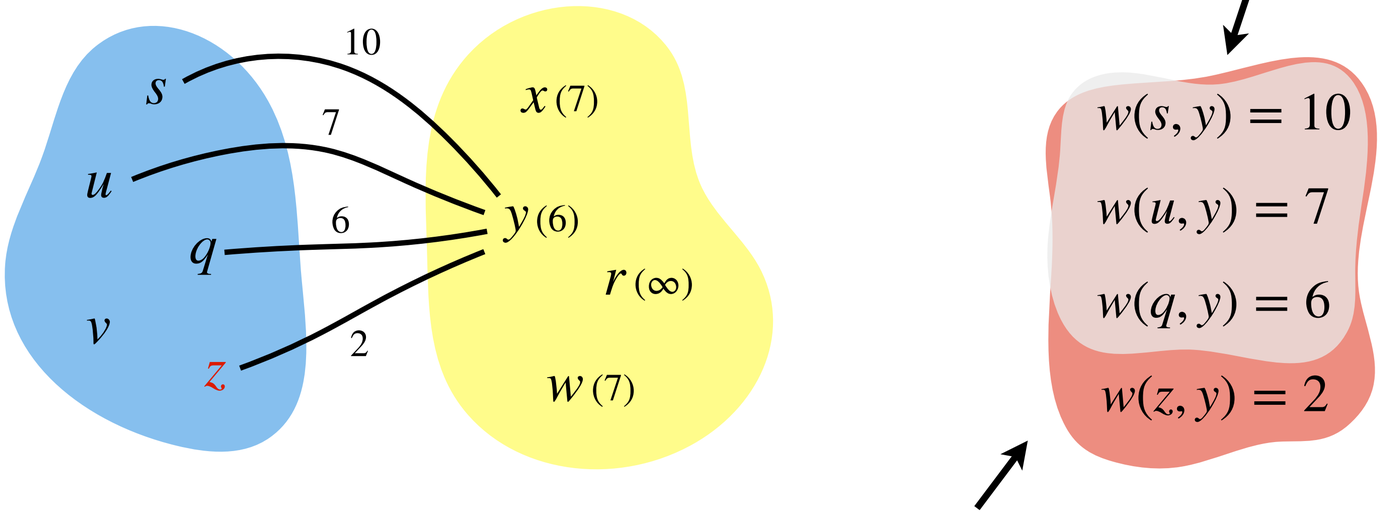

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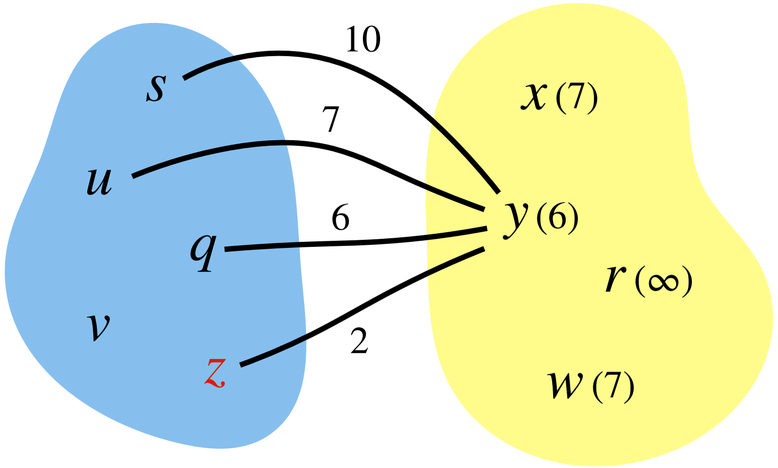

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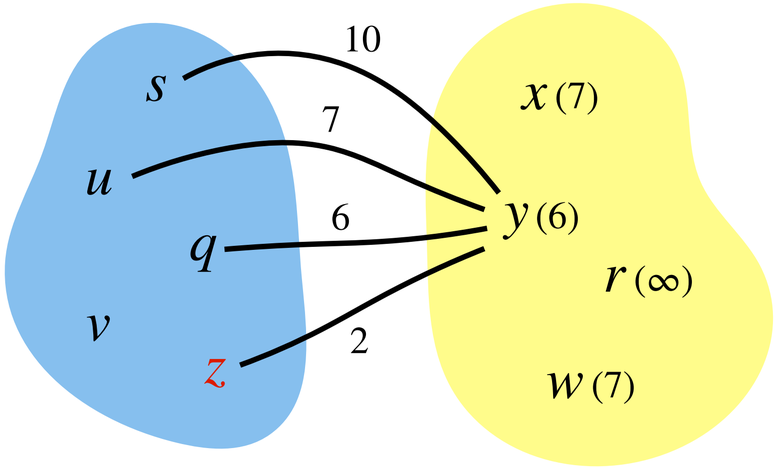

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Min-priority queue/Min-heap.

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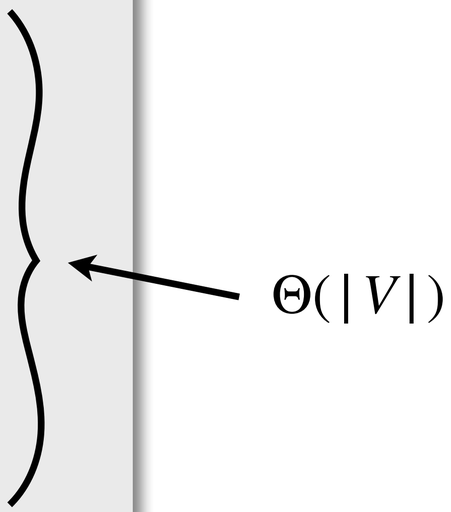

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